Secrecy, Spectrum, and Certification: Evidence from Wireless Electronic Devices*

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Abstract

This paper examines the strategic decisions of firms to withhold information in innovation-intensive markets, highlighting that secrecy contains distinct temporal dimensions. Using all FCC equipment filings from 2001 to 2021, we analyze firms' choices between long-term confidentiality (LTC), which permanently conceals technical information such as block diagrams, and short-term confidentiality (STC), which temporarily delays the public release of marketing information to ensure against launch and coordination risks in the commercialization process. LTC use reflects appropriability concerns and institutional complexities—technically complex products and competitive markets are 12 to 35% more likely to adopt it. By contrast, STC is driven by the commercialization timing of branded products by large firms. Large, domestic, patent-owning firms with regulatorily complex products are more likely to use STC. Technical aspects of the product do not predict STC usage. An exposure-based event study further shows that firms with a high ex-ante propensity to use STC bring products to market faster after policy enactment, with no changes in product complexity or volume.

Keywords: Regulation, Innovation, Disclosure, Spectrum Policy, Intellectual Property

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1 Introduction

Intellectual property rights are a crucial means for firms to capture value in rapidly innovating, invention-intensive markets. Most of the literature focuses on formal exclusivity mechanisms, such as patents, which protect new technologies rather than specific products or inventions. Some studies focus on the use of trade secrets, which pertain to factors within a firm and may apply to multiple products. We investigate a unique setting in which firms have the option to opt into secrecy at the product level due to deregulatory action from a governing body. The setting highlights that secrecy has a temporal dimension. One motive for usage is to preserve "trade secrets," the traditional role of IPR. Another motive is to act as "commercialization insurance," a previously unrecognized role for secrecy to ensure against launch and coordination risks in the commercialization process.

The study leverages regulations indirectly related to the 1996 Telecom Act, when the FCC made new unlicensed spectrum available as part of the U-NII initiative. Much less visibly, it also began to establish a certification system for compliance with FCC regulations. The system authorized third-party firms to perform compliance testing, replacing FCC engineers who had previously performed the testing. In support of this initiative, rules governing device authorization were rewritten and amended. Early in these changes, the FCC began publishing its database of certified products online. Over the next 20 years, the number of new products utilizing unlicensed spectrum increased significantly due to the lower barriers to entry. New opportunities, such as Bluetooth and WiFi (2.4 GHz, 5 GHz, and 6 GHz), also drove continuous technological improvement.

As part of the certification process, the FCC required firms to submit technical and commercial information during the review process. Some of it potentially involved trade secrets, prompting the FCC to define and clarify confidentiality rules. Long-term confidentiality (LTC) had always been available, however, it became adopted once the online database system was established. LTC refers to the permanent protection of specific technical documents submitted by an applicant during the certification process. These documents typically include schematics, block diagrams, operational descriptions, and software security descriptions that contain proprietary information critical to a device's functionality and design. Once granted, LTC ensures that these materials remain undisclosed in FCC public records indefinitely subject to Freedom of Information Act (FOIA) rules.

Short-term confidentiality (STC) was first declared in June 2004 (Federal Communications Commission 2004), with adoption practically starting in 2007. STC provides temporary protection

for certain documents submitted during the certification process, typically for 45, 90, or 180 days after certification is granted. This option is commonly used for external and internal photos, test setup photos, and user manuals—documents that might reveal a product's physical design or functionality before its commercial release for sale to buyers. STC benefits manufacturers who want to avoid premature disclosure of a product's details to competitors or the public before a planned launch. The FCC explicitly states the different goals of the two regimes: LTC is "intended to safeguard trade secrets," while STC is "to allow for the preparation of marketing of devices" (Federal Communications Commission 2011).

Why did some firms and products utilize long-term and short-term confidentiality while others did not? This question attracts our interest because the setting offers a rare opportunity to understand the determinants of keeping information out of public spaces before and after a prospective product launch. The research also benefits from observing the wireless equipment market since the inception of private testing, which yields a longitudinal dataset over the lifetime of the certification program. This enables the tracking of the market's evolution and the identification of different motives behind the use of secrecy, with distinct temporal dimensions, a distinction previously unrecognized in the analysis of firm usage of secrecy.

The study compiles data on every product that underwent FCC review from 2001 to 2021 and documents the use of short-term and long-term confidentiality. The analysis estimates probit models, dividing the analysis into three periods: when long-term confidentiality first diffused, when both long-term and short-term confidentiality simultaneously diffused, and when they reached a plateau. This spans the years 2001-2007, 2008-2010, and 2011-2021. Considering the novelty of the research question, the study adopts a broad approach to measuring the importance of four different factors that shape the use of confidentiality, preferring to test hypotheses with statistical tests rather than rule them out ex ante. The explanatory variables can be grouped into four general categories. The first category reflects competitive pressures at the product level, while the second measures the firm's ability to capture value. A third category measures the role of a firm's experience, and the fourth measures the role of regulatory capture and related regulatory institutions. While the overall certification process had associated costs ranging to the tens of thousands, the marginal cost of requesting confidentiality specifically was negligible for TCB-processed applications. Despite the modest price, we find that not all applications immediately adopted long-term confidentiality when it first became an option. The data show that it took almost a decade for more than 90 percent

of applications to opt for long-term confidentiality, reaching a de facto plateau in 2010 from which it never declined. Relatedly, after it became available in 2007, some firms chose to pay for short-term confidentiality, while others did not. Short-term confidentiality was adopted by just 30% of the applicants within a couple of years. It reached its plateau in 2011.

Findings suggest that the determinants of confidentiality use differ sharply between LTC and STC. For the early period (2001-2007), long-term confidentiality is strongly associated with product complexity and competitive market conditions. The strongest determinant of LTC is the frequency range, which proxies for product complexity. Compared to unintentional radiators (zero frequency ranges), products with one to five frequency ranges are more likely to use LTC by 30 to 40%. Even when considering frequency ranges, products that use Bluetooth and WiFi frequencies are 20% more likely to utilize LTC, and phones are 30% more likely to use LTC. Being in a competitive product market increases the usage probability by an additional 15%. On the other hand, commodity products such as modules and transceivers, which are often outsourced to Asia, decrease LTC usage by 12 to 25%. During this early period, firm institutional knowledge plays a role: Asian firms are 20% less likely to use confidentiality, whereas European and Silicon Valley firms are 10% more likely to do so.

In contrast, short-term confidentiality (2011-2021) is primarily driven by branding and life-cycle considerations. Large North American firms based in Silicon Valley, with products that are regulatorily complex and likely to experience delays, are the most likely to use STC by a total of 45%. In contrast, product complexity, such as frequency ranges, WiFi, and Bluetooth, is not predictive of STC usage. Also, competitive markets are weakly predictive of decreased STC usage. It is still true that outsourcing testing to Asia decreases usage probability by 10%. We interpret this as a result of concerns about product launch delays and the need to protect branding. Interestingly, different factors determine STC compared to LTC, as evidenced by the different margins associated with STC and LTC use.

To address concerns about the changing industry climate from 2001 to 2021, the study also compares LTC and STC in the same period: the transitional time from 2008 to 2010, before steady-state usage in both secrecy policies. Two main results stand out. First, product complexity remains highly predictive of LTC usage (20 to 30% more likely), but it is not predictive at all of STC usage. Second, regulatory complexity is very predictive of STC usage (20% for products with three rule parts) and is not predictive of LTC usage. Accessing the Wi-Fi standard essential patents (SEPs) filed under IEEE 802.11 and those of large members of the Bluetooth SIG, the study then considers

the natural question of how LTC and STC interact with formal IPRs, such as patenting. Findings suggest that patenting and LTC are substitutes, particularly in the case of Wi-Fi during its early period, when it was still an emerging technology. The effect is very different for STC usage, as products with Bluetooth and Wi-Fi SEPs are more likely to use STC.

The study also examines whether policy adoption over time has a different effect on incumbent versus entrant firms, where we define incumbent firms as firms that filed with the FCC before the policy's implementation. Controlling for other factors, LTC take-up is the same between both incumbents and entrants. However, STC take-up is persistently higher for incumbents by 10%, with the difference increasing since the initial policy implementation. We view this evidence as confirmation that LTC primarily involves the protection of technical complexity and the appropriability of product value itself, serving as a substitute for traditional IPR, such as trade secrets or patents. On the other hand, STC is concerned with the protection of branding and commercialization delays, which affect specific types of firms rather than products. To our knowledge, no other study has provided empirical evidence on the subtle differences in secrecy between permanent IP protection and short-term time-to-market protection.

Beyond the cross-sectional determinants of confidentiality, the study asks whether the introduction of STC had a measurable effect on firm behavior. We construct a measure of ex-ante exposure to the STC policy change by using the probit selection equation to predict which firm $\times$ product family cells were most likely to adopt STC based on their pre-period characteristics. An exposure-based event study reveals that higher predicted STC propensity is associated with faster product introductions after the policy becomes available, but not with changes in technical complexity, regulatory complexity, or the volume of products. This finding reinforces the commercialization insurance interpretation: if STC had any measurable behavioral effect, it was on the speed of execution rather than on the nature of the products being introduced.

1.1 Literature Review

The study primarily contributes to the appropriability literature (Levin et al. 1987, Cohen et al. 2000, Mezzanotti and Simcoe 2023), which has identified trade secrets as a crucial strategic option for firms. However, the determinants of using trade secrets are seldom identified in statistical analysis, except as an option when a firm chooses not to patent (Png 2017a,b, Hall and Helmers 2024) or aspires to prevent skilled labor from migrating to competitors (Marx et al. 2009, Agarwal et al. 2009, Ganco et al. 2015, Starr et al. 2021). These alternative studies either examine invention or human

capital. In a rare exception, Cunningham and Kapacinskaite (2026) observe trade secrecy usage at the project level and find evidence linking appropriability regimes to secrecy adoption. The setting at the FCC emphasizes the product and the secrecy surrounding a product's design and marketing, yielding insight into the competitive pressures and value capture issues that shape secrecy at the product level. In this instance, firms opt for trade secrets or make information about a product publicly available, and, importantly for short-term confidentiality, many firms pick either option even after they have shown a willingness to opt into long-term confidentiality. That reveals a previously unexamined temporal dimension of secrecy.

We also contribute to the literature on certifying compliance with government rules. While there is a long-standing interest in examining regulatory rules at agencies that regulate specific industries, such as the FDA, FAA, and SEC (Carpenter et al. 2010, Carpenter and Moss 2013, Stern 2017, Rogers 2023, Chandra et al. 2026), this study sheds light on a practice that has not received much attention: outsourcing compliance activities to third-party inspectors. For example, economists have written about the trade-offs between restaurant inspections by users instead of government employees (Farronato and Zervas 2022). Still, there has been scant writing about outsourcing certification to private entities (Viscusi 1992, Viscusi et al. 2018). Most closely, the study relates to electrical equipment regulation, which is heavily regulated for safety. It is a familiar experience of telephone equipment, where private firms, such as Underwriters Laboratories (UL), conduct most compliance testing for electrical equipment (Bhatia 1984). A related example is the establishment of security frameworks by firm consortia, such as SOC 1 and SOC 2, for cloud computing, where industry organizations conduct certification (Fang and Greenstein 2025). The study contributes by providing a new example for comparison and identifying the benefits of privatizing certification.

This work contributes to the analysis of the regulation that supports devices using licensed and unlicensed spectrum (e.g., Milgrom et al. (2011), Hazlett (2017), Fontana and Greenstein (2021), Kim et al. (2024) and several other studies), as well as various devices that interact with licensed spectrum. We provide an analysis of the confidentiality provisions, which were part of many reforms initiated by the FCC after the 1996 Telecom Act. A notable paper in this literature is Trlifaj (2025), who studies the release of the FCC database in 1998, which focuses on the government rules that require transparency. In contrast, we focus on later events and other aspects, while also touching on the same regulatory setting. As part of this regulatory analysis, we examine whether established firms exploit the options for confidentiality for their private benefit. Standard capture theory suggests

that regulatory changes favoring firms (such as granting them secrecy) are a giveaway to established firms, protecting them from competition (Stigler 1971, Carpenter and Moss 2013). We also examine whether, as speculated by Hazlett (2017), the unlicensed regime enables many equipment providers to experiment and find value from new product designs. Experimentation requires many suppliers and rapid product introductions, and the regulatory rules that support such experimentation in markets need to be recognized and assessed. Secrecy is a part of that because it helps the novel design slow down the imitator.

Finally, equipment that utilizes licensed and unlicensed spectrum has become a multibillion-dollar market but has received less attention than the policies governing the allocation of licensed spectrum. The U-NII was a prominent initiative aimed at allocating spectrum to unlicensed users. Devices that utilize unlicensed spectrum have received less attention than the supply chain for licensed spectrum in cellular networks, and this study begins to address that gap.

The remainder of this paper proceeds as follows. Section 2 describes the institutional context, and Section 3 discusses the underlying theory and framework. Section 4 describes the data construction and Section 5 documents descriptive evidence and the analysis methodology. Section 6 presents the selection equation results, Section 7 examines whether STC exposure affected firm behavior using an event study design, and Section 8 concludes.

2 Institutional Setting and Regulatory Changes in Unlicensed Spectrum

Like its predecessors, the 1996 Telecom Act requires that every device using the electromagnetic spectrum in the United States be authorized to ensure it does not interfere with other users of the spectrum. In practice, this requires the manufacturer to demonstrate compliance with restrictions on power and band usage, as well as other technical requirements.

Up until the late 1990s, the FCC employed more than 30 engineers to test for compliance. The system of certification for interference came under strain during the early growth of the commercial internet and consumer electronics in the mid-to-late 1990s. This testing and certification process proved too small for the number of new products requiring certification. As a first reform, the FCC eliminated products that fell into the compliance system but posed no real danger of interference because their emissions had a small geographic reach (Knapp and Wall 1997). It began to classify them as "unintentional radiators."[1] As it turned out, this modest change did reduce the number of required certifications for a short period, but ultimately did not resolve the issues caused by capacity

[1] Unintentional radiators unintentionally emit RF energy (e.g., computers and power supplies).

limitations. Many of the same products (such as printers) began to use wireless features, bringing them back into the certification process. Delays were particularly problematic for firms with new products that have short life cycles. Delays of three to six months could have considerable adverse consequences for commercial prospects.

One other factor played a role. By the late 1990s, experiments with unlicensed spectrum – on baby monitors, garage door openers, and mobile handsets – demonstrated that regulations on power and emissions could lead to products that utilize unlicensed spectrum for short-range applications. Researchers had organized additional experiments with prototypes for various applications, such as campus wireless support and point-of-sale systems. These prototypes raised the prospect that a broad range of applications could utilize more unlicensed spectrum. Following the passage of the 1996 Telecom Act, several changes were implemented to encourage greater use of unlicensed spectrum. The addition of the spectrum band in the 5 GHz band was authorized as the U-NII bands.[2] This was added to the 2.4 GHz band for consumer devices in the unlicensed spectrum.[3] In addition, efforts to define protocols at the IEEE Committee 802.11 (later Wi-Fi) and Bluetooth Consortium accelerated and began to spread to many products in the early 2000s (Fontana and Greenstein 2021).

In addition, the FCC began implementing a system for testing compliance with testing and certification from authorized private firms. In 1998, it adopted rules to permit Telecommunications Certification Bodies (TCB), the first of which was approved in June 2000. The TCBs seek compliance with emission, power limits, and interference rules. A device must be sent to an FCC-accredited laboratory, which may be independent of any TCB. After certification of compliance, the TCB enters the device into a public record.

Starting in 2001, this led to a new cost structure and timeline for obtaining FCC approval. Firms with new devices hired TCBs, who had the option to test the devices with their staff and equipment or hire another testing firm. Once a product had passed the tests, the TCB applied for certification on behalf of the firm, entering the device into the public database, where it is assigned an FCC ID.[4] Generally, testing took one to two weeks, and TCBs' review and approval also took one to two weeks. The entire process, if successful, could be done in a few weeks. The expenses were modest, typically ranging from a few thousand dollars to a couple of thousand dollars.[5] A summary

[2]Unlicensed National Information Infrastructure.

[3]Other bands of the unlicensed spectrum had been reserved for industrial, scientific, and medical purposes.

[4]The FCCID is created by the manufacturer by combining a grantee code (assigned by the FCC to the manufacturer) and a product ID (chosen by the manufacturer).

[5]Evidence on historical costs is challenging to find. Today, fees vary. For a simple Wi-Fi or Bluetooth device,

of this system is presented in Figure 1.

The FCC maintains a public record, which began transitioning to an electronic format in the early 1980s. It was first made available online in 1998 following changes to streamline the equipment authorization process in a 1997 IEEE conference (Knapp and Wall 1997). Long-term confidentiality (LTC) rules started to be widely adopted in 2001, just after the TCB and fully electronic filing systems were implemented. Typical exhibits which can be held permanently confidential include schematics, block diagrams, operational descriptions, and parts lists. After many years of experimentation with this new set of rules, the short-term confidentiality (STC) rules appeared officially in 2004, and practically in 2007.[6] Typical STC exhibits include external photos, internal photos, test setup photos, and user manuals. This allowed commercially-sensitive information to remain confidential until the actual marketing of the device (Federal Communications Commission 2011). Table 1 details the confidentiality rules around each type of exhibit.

The fees tied to confidentiality applications were mostly administrative: ensuring there is an explanation with the request and separating out the exhibits. Due to the one-time nature and pervasiveness of LTC, many TCBs started offering LTC as part of the certification process at no extra cost (American Certification Body, Inc. 2018). On the other hand, STC evoked an additional fee since it was subject to more complex rules and thus required monitoring. STC can be granted for a maximum of 180 days, or whenever the product is first marketed or launched. If the firm does not know the latter at the time of application, 45 days can be requested with additional extensions possible. All STC exhibits must be made public at the first moment of device publicity to avoid a breach of FCC policies (American Certification Body, Inc. 2018).

The commercial environment was also changing at the same time. Private firm consortia were designing new protocols to use the unlicensed spectrum simultaneously and outside the FCC's control. This included competing designs for wireless local area networks from HomeRF and the IEEE 802.11 committee, as well as short-range applications from the Bluetooth consortium, among other initiatives. The 802.11 committee's standard eventually prevailed due to the actions of private firms involved in the laptop supply chains (e.g., Fontana and Greenstein (2021)). The FCC does not possess the legal authority to enforce conformance with IEEE 802.11, which was later renamed

the costs are modest: \$1,500-\$3,500 for TCB certification and \$5,000-\$15,000 for testing costs. For a router: \$10,000-\$25,000 for TCB certification and \$3,000-\$6,000 for testing. For a phone: \$25,000-\$50,000 for TCB certification and \$5,000-\$10,000 for testing.

[6] Once the product was in the FCC database, anyone can challenge the grant, including the grant of confidentiality. However, challenges were fairly rare.

Wi-Fi after its widespread adoption. That left the private consortia in a position to enforce the standard, a responsibility it assumed. The Wi-Fi consortium continues to certify compliance with Wi-Fi protocols.

As the data below shows, after the regulatory rule changes, devices that met the Part 15 regulations became the largest category of devices. The FCC Part 15 rules govern the operation of unlicensed radio frequency (RF) devices in the United States, ensuring they do not cause harmful interference to licensed communications. These rules apply to various electronic devices, including Wi-Fi routers, Bluetooth devices, RFID systems, and cordless phones. Part 15 categorizes devices into two types: intentional radiators, which intentionally emit RF energy (e.g., Wi-Fi transmitters), and unintentional radiators, which unintentionally emit RF energy (e.g., computers and power supplies). While the rules in Part 15 cover devices that use unlicensed spectrum, so do Parts 18, 95, 96, and 97. For licensed spectrum, different Parts apply. Many cellular and smartphone devices follow the rules defined in Parts 20, 22, 24, 27, 90, and 96 of the FCC regulations. The FCC's rule changes also affected additional Parts, including Parts 21, 73, 74, 80, and 9, which are not the focus of this study. See Appendix A1 for further explanation.

3 Theory and Framework

In this section, we review the various streams of literature that motivate our empirical analysis and relate them to the specific context of confidentiality in the case of unlicensed spectrum.

The first stream of literature we consider concerns firms' incentives to disclose information when competing based on product quality. The original result in this literature goes back to the 'unravelling prediction' (Grossman and Hart 1980, Grossman 1981, Milgrom 1981) that states that a firm will voluntarily disclose information as long as it is costless to do so, the intuition being that, if the quality of the product is random and not observable, customers would be willing to pay only the expected value. Hence, the incentive for firms with higher-quality products is to disclose information, thereby avoiding customers from revising their expectations downward and allowing them to charge a higher price. Further research in this stream of literature has explored the conditions that may lead firms to withdraw information. Grossman and Hart (1980) have shown that disclosure might be limited to high-quality products if disclosure is costly. Other scholars have investigated the relationship between competition and the incentives for firms to disclose information on their products. In this respect, the evidence is mixed.

On the one hand, some argue that intense competition leads firms to withhold information on

production costs and demand that might further reduce price margins (Okuno-Fujiwara et al. 1990). This result has been corroborated by Jin (2005), who, in the context of the healthcare industry, shows that disclosure rates tend to be lower in highly competitive markets. However, she also argues that disclosure depends on product quality and that quality differentiation can be used to shield products from competitors. Board (2009) shows that the fear of increased competition may lead firms to withhold information. In his model, the decision of (some) high-quality firms to disclose changes the incentives for rival firms to do the same, as they have to trade off the increase in competition if they disclose with the reduced (customer) expectations if they do not. It also has a positive effect on low-quality firms, allowing them to capitalize on rising customer expectations without disclosure. On the other hand, researchers have also argued that increasing competition may instead sharpen the incentives for revealing information. This may happen because, under intense competition, even small changes in quality perceptions become important (Stivers 2004) or because, when the market is unaware that the firms possess any new information, the announcement by one firm of news acts as a signal to the market that other firms may also possess new information (Dye and Sridhar 1995).

An extensive literature has investigated the effects on a firm's competitive expectations following a change in competition in the same product market (Jansen 2010, Bloom et al. 2013, Cookson 2018). Competition from a rival firm's product leads to increasing concerns about reduced consumer interest, as well as an increase in the strategic significance of potential information spillover, which may prompt the focal firm to withdraw information on its own products and/or R&D projects. In the context of the pharmaceutical industry, for instance, Kao (2025) demonstrates that increased competition from rival drugs decreases the likelihood of disclosing the results of clinical trials.

Further implications for the relationship between incentives to disclose information about new products and value capture can be drawn indirectly from the literature on product portfolios. Firms with broad product portfolios operate in multiple market segments and often have more at stake in terms of reputation (Cabral 2000), brand signaling (Pepall and Richards 2002), and market expectations. Strong brand enables firms to capture higher price premia and enjoy higher economies of scope because relatively less is spent on advertising (De Figueiredo and Kyle 2006). Therefore, firms may have stronger incentives to disclose innovations when introducing new products in a previously unexplored segment. These incentives might also be motivated by the need to signal overall innovative capability across the portfolio. They might be aimed at shaping expectations among investors, partners (Bushee et al. 2024), or regulators. Alternatively, they may seek to preempt competitors

by signaling and setting higher quality standards (Baker and Mezzetti 2005).[7] Additionally, broad portfolios can help mitigate the risks associated with disclosing sensitive information. If a disclosure backfires due to imitation or customer dissatisfaction, it may be less consequential when the firm's value is not tied to a single product line. This encourages greater openness about innovation. The flip side of having broad product portfolios is the risk of cannibalization. Firms with broader portfolios might delay disclosure if new products cannibalize existing ones (Moorthy and Png 1992). This is especially true in the consumer electronics and pharmaceutical industries (Chu et al. 2025), where the introduction of new products can reduce the lifecycle of existing ones.

The second stream of literature we consider is about innovation and appropriability. It focuses on the use of secrecy to appropriate returns from innovation, and as an alternative to patenting (see, among others: Koo et al. (2025), Hall and Helmers (2024), Crass et al. (2019), Hall et al. (2014)). Surveys of innovative firms that compared the effectiveness of patents with other appropriability mechanisms found that firms tend to prefer trade secrecy over patents to secure their R&D efforts (Levin et al. 1987, Cohen et al. 2000, Mezzanotti and Simcoe 2023).[8] Similar to patents, certification processes at the FCC disclose information on product characteristics, underlying technologies, and product designs, thereby exposing them to imitation by rival firms. Contrary to patents, however, certification does not provide firms with protection against imitation.[9] The choice of secrecy over patenting has been associated with process innovation or with complex products that have a substantial component of tacit knowledge, which is difficult to codify into a patent. When the innovation is embodied in a 'process', like in the manufacturing of chemicals, or it is broken down into pieces not clearly identifiable, as in the case of 'source' and 'object' code in software, knowledge cannot be easily accessed and imitated (Graham 2004). Survey-based evidence (Arundel 2001, Cassiman and

[7] Baker and Mezzetti (2005) argue that through selective (patent) disclosure, firms can enlarge the scope of the prior art, raise the threshold of patentability, and thus preempt competitors from patenting. Disclosure is seen as a way of extending the patent race, as, after the disclosure, the invention should represent a sufficient advance over an expanded prior art. If the disclosing firm is a leader in the patent race, extending the threshold may discourage the rival from catching up. If the disclosing firm is lagging, extending the race might give more time to catch up.

[8] Patents disclose important information on innovation characteristics, technology, and product design which might be imitated by competitors or used to invent around or reverse engineering. Hence, there is a need to ensure protection and provide the patent holder with the proper incentive to undertake the innovative effort in the first place. By choosing secrecy over patenting, an innovator does not disclose information about the innovation, thereby lowering the risk of being copied by rival firms.

[9] An extensive literature has analyzed the positive effect of innovation disclosure on invention and job creation (Furman et al. 2021), high technology licensing (Mohammadi and Khashabi 2017), corporate venture investments (Hegde and Luo 2018), R&D investments (Hegde et al. 2023), and follow-on innovation (Bertolotti 2026). Certification, like disclosure, should lead to more product introductions (Zhang et al. 2026). However, it also increases competition in the marketplace and, therefore, the propensity for firms to apply for confidentiality. This is the aspect we are interested in investigating in this paper.

Veugelers 2002) highlights that the complexity of product design is one of the most popular means of protection chosen by firms that put efforts into designing increasingly complex products to make imitation by rival firms time-consuming and difficult (Markopoulos and Hosanagar 2018, Ethiraj et al. 2008).

Other contributions have argued that the choice of specific appropriability mechanism (i.e., patents or secrecy) may depend on firms' business characteristics, the institutional environment (Gross 2023, Hou et al. 2023, Png 2017a,b), as well as competitive strategies. In this respect, the predictions are mixed. Some papers have investigated the use of patents by incumbent firms to deter entry. Bunch and Smiley (1992) used survey data to find early empirical evidence on the use of patents to prevent entry. In semiconductors, Ziedonis (2004) found that firms expand their own portfolios of patents to avoid being 'fenced in' by owners of patented technologies. Ceccagnoli (2009) demonstrated that incumbents employed preemptive patenting to enhance their market power. This applied in particular to incumbents with higher ex ante market power and facing a higher threat of entry. In their analysis of the software industry, Cockburn and MacGarvie (2011) found evidence consistent with the idea that patents have an entry-deterring effect in addition to reflecting the technological capabilities of the firms that generate them. More recently, in the context of the pharmaceutical industry, Conti et al. (2025) have demonstrated that, in response to a competitor's entry, incumbents intentionally fragment and delay the full disclosure of their intellectual property rights through the use of continuation patents. When entry concretizes, incumbents reveal their continuation patents to delay competitors through litigation. Their reaction is stronger the more valuable are the drugs.

Another set of papers has investigated the choice of secrecy. Anton and Yao (2004) built a signaling model to study the strategic disclosure of information through patenting. They found that small innovations are fully revealed, but large innovations are mainly kept secret. Sim (2021) demonstrates that for complex innovations (i.e., innovations composed of complementary knowledge components), patenting and secrecy can be used in conjunction to extend the market power of an incumbent firm and provide better protection than patenting alone. Schneider (2008) and Zaby (2010) consider the relationships between technological leadership and the choice of secrecy. They show that if a firm has a technological lead over a rival, it is more likely to utilize its leadership in generating new products. However, Gill (2008) demonstrates that a firm with a significant lead over its competitors may choose not to utilize secrecy to deter them from competing. Thus, the larger the lead, the lower the probability of applying for secrecy. In general, leaders might prevent rival laggards

from catching up by pushing the technological frontier forward radically or by frequently innovating incrementally. If leaders innovate frequently by modifying the characteristics of existing products, they can maintain their leadership over rivals independently of seeking secrecy (Zhang 2012).

Another relevant stream of literature concerns regulatory settings and new product introductions. Regulation may affect the costs and requirements of introducing new products to market, and therefore is expected to influence both the timing and the efforts to innovate. Prior studies, particularly in the pharmaceutical and medical device sectors, have corroborated these expectations.

A key issue that this literature has examined is the impact of prior experience with regulation on shaping firms' product introduction strategies. In the context of drug review, Carpenter et al. (2010) have shown that first-mover firms in a market niche are more likely than later entrants to receive approval, even when the quality of their products is lower. They also pointed to 'consumer cooptation' by the regulator as a possible explanation of this result.

In the context of medical devices, Stern (2017) has found that the approval time for pioneer firms in a new category of device is longer than that of follow-on innovators in the same category. She explains this result based on 'content and format' rather than technological uncertainty—(Classes of) products that lack a regulatory history or established regulatory criteria experience longer approval times. Less experienced (i.e., small and more financially constrained) firms are less likely than big firms to enter markets where regulatory uncertainty is present.

Prior experience may lead firms to respond differently to changes in regulation stringency. In the medical device industry, Rogers (2023) has found that declassification of the risk regime has favored innovation in certain types of medical devices. In particular, innovation was highest among the smallest firms and those with limited regulatory experience. Additionally, deregulation led to changes in competition, as both new firms and, to a lesser extent, existing incumbents entered the market.

A related aspect is the possibility that more experienced firms may influence the regulator in their favor and/or at the expense of competitors due to a 'strong' or 'weak' form of capture. Capture theory in economics explains how regulatory agencies may become dominated by the very industries or interests they are supposed to regulate. Early contributions in this strand of literature explain capture as a consequence of political actions (Stigler 1975, Peltzman 1976, Viscusi 1992). Large and more resourceful firms can better organize politically, influence politicians and regulators, and build entry barriers against smaller and less experienced competitors. More recent contributions

have shown that larger and more experienced firms may gain advantages even in the presence of a 'neutral' regulator. In the context of the pharmaceutical industry, Carpenter (2004) identifies several sources of advantages for large firms. One source is the familiarity with the regulators and the regulatory framework in general. As they introduce products more frequently, the quality of large firms' products is better known to the regulator, thus leading to faster approval. Another source is lower opportunity costs with respect to small competitors. As they own bigger assets, large firms can better absorb the consequences of delays and negative outcomes in the approval process than smaller firms. Finally, thanks to their capitalization, large firms enter market niches early and benefit from quicker approval.[10]

Taken together, these streams of research point to the following determinants of confidentiality and associated predictions in the empirical setting of this study.

Regulatory Advantages. A set of determinants measures a firm's familiarity with regulatory rules and risks. In some cases, maintaining confidentiality can help firms avoid certain legal or regulatory hurdles. In other instances, revealing little about a product might also make it less vulnerable to litigation. An advantage may accrue to firms with long-standing relationships with regulatory authorities, regardless of their country of origin. These factors would be especially salient immediately after the unveiling of new regulatory rules. A strong form of capture theory would predict that firms with more experience in the regulatory setting will maintain robust conversations with regulators and utilize the regulatory process to put their rivals at a disadvantage. This observation leads to two distinct predictions. If US regulators favor US firms with more information and more notice about rules, then US-based firms would utilize confidentiality rules more frequently than otherwise similar firms from outside the US.

Prior Experience. We expect established and experienced firms to utilize short-term and long-term confidentiality at higher rates than entrepreneurial firms and those with less experience in the communications equipment sector. Alongside regulatory rules, this experience can also extend to market advantages. The literature on patents and secrecy has suggested that established firms may utilize confidentiality to build upon an existing first-mover advantage. The existing literature does not support an unambiguous prediction, as applying for secrecy seems to depend on the distance between leaders and laggards.

[10] Carpenter (2002) and Kyle (2002) find that entry probability increases with firms' market experience and decreases with the number of previous entrants. Olson (1997) finds that larger firms (in terms of employment) and firms with more products have shorter approval times.

Additionally, although confidentiality works similarly to secrecy in that it denies competitors access to information, it differs from secrecy in several key ways. The unit of analysis differs. Specifically, confidentiality applies to products, not to inventions themselves. As many studies have shown (see, for instance, de Rassenfosse and Zhou (2020)), it is challenging to establish a one-to-one correspondence between products and inventions. Relatedly, this difference enables firms to opt into confidentiality for some products and not others.

Additionally, our setting has a temporal dimension that is not present in the literature on patents or trade secrets. After 2007, the FCC allowed firms to choose between long-term (LT) and short-term (ST) confidentiality. The former applies to designs, diagrams, and code. The latter applies mainly to user manuals, which are released when the product is launched. LT confidentiality allows perpetual protection. The benefits of LT confidentiality may be extensive, ranging from having time to manage novel technologies and gain competitive advantages over rivals, to designing complementary products, and to developing the capabilities necessary to commercialize the innovation successfully. When ST confidentiality expires, the codified elements of know-how will be disclosed, allowing for reverse engineering and other commercial responses.

These features, at the product level and with options for both short-term and long-term use, will interplay with firm advantages developed in specific equipment markets. Long-time participants in the equipment market with familiar brands will bring valuable experience to the design, distribution, and marketing of equipment. This will convey advantages over new entrants, who typically possess less experience and less well-known brands. This experience may also extend to understanding and employing experienced engineers who know how to design products that pass certification.

Value Capture. By requesting confidentiality, firms can gain a lead in bringing a product to market, allowing them to establish a strong market presence and brand recognition before competitors can respond. Following this logic, we would expect firms that launch products that are 'new to the market' to be more likely to apply for confidentiality than firms that launch products 'new to the firm'.

By maintaining confidentiality, a company can strategically control the timing of product releases to minimize overlap and protect the returns on its current sales. When a firm has multiple products, the disclosure of new features or enhancements may lead to internal competition with its existing products. This internal competition occurs when a new product's sales diminish the sales of the company's existing products (i.e., market cannibalization). We would expect firms with 'deep'

(i.e., many products in one or a few segments) rather than 'broad' product portfolios (i.e., many segments with not necessarily many products) to be more likely to seek confidentiality.

Relatedly, confidentiality enables a company to craft a precise commercialization strategy, including the development of targeted marketing messages and the effective integration of new products into its brand. This might be particularly important for incumbent firms with an established reputation and brand, which are concerned that new products target distinct consumers while maintaining the brand's appeal for existing ones.[11]

Competitive Pressures. In the context of this study, getting products to market after certification can take time, ranging from a few weeks to several months. Keeping product design and features confidential for a short period prevents competitors from gaining insight into a company's product pipeline. Confidentiality ensures that competitors cannot preemptively launch similar products, which could saturate the market and depress the demand for the focal firm's future products. The same applies to long-term confidentiality. By not disclosing proprietary information, a firm maintains its competitive edge. We would expect this motivation to be more relevant in markets where competition is intense and/or where leaders have a limited lead over laggards.

In addition, withholding information from rivals may help the focal firm benefit from prior investments and explore new product features, while preventing competitors from quickly imitating or learning from past failures. Both short-term and long-term confidentiality may help firms differentiate their products both horizontally and vertically from those of their competitors, enabling them to charge premium prices for their offerings. Without competitors having access to the same knowledge, firms can avoid price wars and increase their profit margins. We expect more complex products (in terms of the number of technical characteristics) and those of relatively higher quality to provide the highest economic benefits from differentiation and, therefore, to be more likely to utilize options for confidentiality. (For definitions of complex products, see Section 4.3).

Finally, in the context of product certification, short-term confidentiality offers additional benefits in the form of insurance against the possibility that something goes 'wrong' during the certification process, potentially delaying plans for manufacturing or distribution of the product. Firms employ short-term confidentiality to avoid revealing the details of their new products to both

[11] To the extent that product innovation frequency depends on competition, an increase in the number of rivals may be expected to lead to less confidentiality. Ponce (2007) shows that an innovator may opt for secrecy but will disclose some knowledge to prevent a potential second innovator from developing the same innovation and patenting it. However, this insight pertains to inventions, so it is unclear whether it will apply to products.

competitors and customers before a delayed product launch.

There are many overlapping concerns between these last two categories. For example, many firms outsourced their product assembly to save costs and respond to competition from others who also outsourced their products. We have an ambiguous prediction for outsourcing. We would expect these pressures to make firms more likely to want to protect the value of their products until the point that they become commodities with no margin, at which point the value of outsourcing is diminished.

Our theoretical framework explains the decision to apply for confidentiality as a function of a series of variables grouped into four broad areas. These variables are not mutually exclusive and can be translated into measurable proxies for product-level and firm-level analysis. Table 2 presents a list of our proxies, along with some predictions.

4 Data

This section explains the data used to implement the conceptual categories outlined in Table 2. Table 3 computes the average value of each variable across different sample time periods.

4.1 Overview of FCCID.IO

The FCC maintains a database of all products using unlicensed spectrum that apply for certification, which is replicated in a more data-friendly format at a third-party website called fccid.io. We scrape all data from 1982 to 2021. Since confidentiality policies were implemented along with TCBs due to the public disclosure of exhibits, we restrict the sample to 2001 and onwards. All analyses will be restricted to products in Parts 15, 18, 22, 24, 27, 81, 87, 90, and 95, which constitute over 95% of all products in the dataset (Table 4). Parts 21, 73, 74, 80, and 9 are excluded since they include broadcasting equipment that differs significantly from most unlicensed spectrum devices.

The dataset is unique at the application level, not the product level (identified by a fccid). The same fccid can file for subsequent applications for various reasons, most of which pertain to product modifications known as a "Class II" or "Class III permissive change." For the purposes of this study, we restrict the sample to "Original Equipment" applications at the first granted date. Now we can perform all analyses at the product level.

Each fccid has its own page on the website and includes information on the applicant firm, the certification and testing firms, a brief product description, application and grant dates, operating frequency ranges, and confidentiality usage. All exhibits (e.g., User Manual, Internal/External Photos, Testing Reports, etc.) are also included on the website. However, since they are uploaded as separate PDF files, they are not part of the scraped dataset.

4.2 Firm Identifiers

The first three to five characters of the fccid correspond to a "grantee code", which corresponds to a unique applicant firm. However, the same firm can have multiple grantee codes that change over time. For example, Apple has grantee codes "BCG" and "2AN3R" which were registered in 1998 and 2017, respectively. To accurately assign products to firms, we implement a clustering algorithm on available firm information to generate our own unique firm IDs. We group together firm names, addresses, and grantee codes and use the Python package `disamby` to generate clusters of disambiguated firms. This generates a total of over 32,000 disambiguated firms in our sample. However, there are cases where different firms with similar names are clustered together and the same firm with different subsidiary names are not clustered together. We manually inspect major firms to fix such cases.

4.3 Complexity Measures

The primary technical information for each product is the operating frequency range. Around 3.6% of products are unintentional radiators and have zero frequency ranges. Around two-thirds of products in the sample have a single frequency range. We use the number of frequency ranges to proxy for a product's complexity. More complex products are certified to operate in multiple frequency ranges. The FCC also identifies operating frequencies that fall under NFC (near-field communication), Bluetooth, 2.4 GHz Wi-Fi, 5 GHz Wi-Fi, and 6 GHz Wi-Fi. Due to the limited number of products that operate NFC and 6 GHz Wi-Fi, we do not include these variables in our analysis.

4.4 Product Markets

Due to the increase in exhibits following the TCB reform, most products submitted after 2000 include a short description. To classify products by target market, we first run an unsupervised clustering algorithm to group the product descriptions into 80 clusters (the optimal number based on manual inspection). The top repeated phrases are extracted from each cluster to build a key-phrase dictionary (KPD), along with additional phrases identified through manual inspection. All product descriptions are matched to entries in the KPD.

Each of the 4261 key phrases is manually mapped into product markets inspired by NAICS codes and the 2022 Consumer Technology Association report. We build three levels of product markets that nest within each other: 26 Classes, 83 Families, and 298 Types. Note that while we keep the NAICS codes as a reference, they are far too broad for our purposes. Each product market is also assigned as either an "end market" (B2C) or an "intermediate market" (B2B). The full list of product markets are provided in Table A.1. See Greenstein et al. (2026) for an in-depth discussion

of the constructed FCC product markets.

4.5 Competition Measures

For each of the three product market levels, we compute the number of products in that market per year, weighted by firm "survival percent". The survival percentage is a proxy for the firm's product quality, assuming that firms who survive longer produce higher-quality outputs. It is a firm-level measure that computes the percentage of time since the first product grant date until the last date in the dataset that a firm applies for products:

$$\frac{\text{(Last Product Grant Date + One Year) – First Product Grant Date}}{\text{Last Date – First Product Grant Date}}$$

We assume that firms who do not apply within a year have exited the market, hence we add one year onto the last product grant date. We truncate all survival percentages at 100%. Once we have product market times year-level measures for "competition", we split them into those above and below the median competitiveness.

4.6 Patent Measures

We merge data on Standard Essential Patents (SEP) to evaluate the level of substitutability between confidentiality and traditional forms of IPR. We merge all firm-level Letters of Assurance (LOA) under the IEEE Wi-Fi standard 802.11. To capture Bluetooth SEPs, we merge data on firms that are main contributors to, or promoters of, the Bluetooth Special Interest Group (SIG), or the standard 802.15.1. There are two types of variables we can construct. First, we construct firm-level variables, such as whether the firm is an SEP holder, whether it owns many Wi-Fi LOAs, or whether it chooses to disclose its patents in LOAs. Second, for each product, we observe in the FCC database whether they were tested for Wi-Fi and/or Bluetooth frequency ranges.

4.7 Testing and Certification

The FCC database provides the TCB and the testing firm that certified and tested each product, respectively. TCB information includes the name, address, and contact email. We can match all TCBs to the FCC list of certified TCBs (42 as of 2022) or to a TCB merger. Testing firm information includes only the name and contact email. However, the name of the testing firm usually consists of the city, such as "Shenzhen United Testing Technology" or "Bay Area Compliance Laboratories (Dongguan)". We can classify most of the testing firms' locations as Asian, European, US, or Other based solely on their names. For a few testing firms with undetermined locales, we manually check the testing report exhibits. Then, based on the applicant firm's location, which is part of the scraped database, we can determine whether North American or European firms decide to outsource testing

to Asian firms.

5 Descriptive Evidence and Methodology

Figure 4 reveals two striking descriptive patterns. First, the take-up of long-term confidentiality diffuses slowly over a period of seven years, rising from 40% in 2001 to over 80% by 2008. Second, short-term confidentiality is adopted rapidly after its introduction in 2007, with usage stabilizing at approximately 30%. A natural question arises: despite the low cost of confidentiality relative to the overall certification process, why do not all products ultimately opt for short-term confidentiality, and why is adoption not immediate for long-term confidentiality? The theory outlined in Section 3 suggests that confidentiality usage should vary systematically with competitive pressures, value capture incentives, prior experience, and regulatory familiarity. If these factors operate differently across firms and products—and if long-term and short-term confidentiality serve distinct strategic purposes—then the observed heterogeneity in take-up is precisely what we would expect.

Given this dynamic and cross-sectional heterogeneity, we analyze confidentiality usage separately across three time periods that correspond to distinct institutional regimes. The first period, 2001 to 2007, follows the TCB reform and introduces the option of long-term confidentiality during certification. By opting in, firms permanently withhold technically sensitive materials such as schematics and block diagrams—the type of proprietary information that the appropriability literature identifies as central to sustaining competitive advantage in innovation-intensive markets (Levin et al. 1987, Cohen et al. 2000). This early period is marked by the rapid entry of Asian firms following China's accession to the WTO in 2001, and by the slow diffusion of long-term confidentiality. Short-term confidentiality has not yet been enacted.

The second period, 2008 to 2010, begins when the FCC implements short-term confidentiality in response to industry demand. Short-term confidentiality protects marketing-related information—such as user manuals, product photos, and test setup images—during the interval between certification and commercial launch. As discussed in Section 3, this form of protection addresses a distinct concern: not the permanent concealment of trade secrets, but insurance against the coordination risks that arise when the timing of regulatory approval and product launch diverge. During this transition period, short-term confidentiality diffuses rapidly, while long-term confidentiality usage continues to climb.

The third period, 2011 to 2021, represents a steady state in which usage stabilizes at 95% and 30% for long-term and short-term confidentiality, respectively. The near-universal adoption of

permanent IP protection (LTC) contrasts with the selective take-up of commercialization protection (STC). This divergence is consistent with the theoretical distinction between appropriability-driven secrecy, which all firms eventually find valuable as certification databases become more accessible, and timing-driven secrecy, which matters primarily for firms with specific branding and launch coordination needs. Because LTC adoption has largely saturated by this period, the analysis of the late period focuses exclusively on the determinants of short-term confidentiality.

There are two possible levels of analysis: firms may determine confidentiality on a product-by-product basis, or they may implement firm-wide policies. Tables 5 through 7 cross-tabulate long-term and short-term confidentiality usage for products and firms in 2011, a representative late-period year. Two facts emerge. First, virtually all products that opt into short-term confidentiality also opt into long-term confidentiality, confirming that STC is adopted as a complement to, rather than a substitute for, permanent protection. Second, a substantial share of firms that use any level of confidentiality do so for only some of their products, rather than uniformly. Confidentiality usage thus varies *within* firms, and the relevant unit of analysis is the product rather than the firm. This within-firm heterogeneity is consistent with the theoretical prediction that competitive pressures and value capture incentives operate at the product level—a firm may have strong reasons to protect a technically complex flagship product while leaving a commodity module uncovered.

5.1 Methodology

What drives the heterogeneous take-up of different confidentiality regimes? The theoretical framework in Section 3 identifies four conceptually distinct categories of determinants. The first two—*regulatory setting* and *firm experience*—pertain to institutional factors, such as regulatory familiarity, capture, and accumulated organizational knowledge. The remaining two—*value capture* and *competitive pressures*—pertain to commercial factors, including appropriability concerns, market structure, branding, and the risk of information spillovers. These four categories generate a rich set of predictions, some of which differ for long-term and short-term confidentiality. For example, the appropriability literature predicts that technically complex products in competitive markets should be more likely to adopt permanent secrecy, whereas the literature on product portfolios and commercialization timing suggests that large, brand-oriented firms should be more likely to adopt temporary protection. The regulatory capture literature, by contrast, predicts that experienced domestic firms should enjoy advantages in both.

We organize the empirical analysis around three questions that map directly onto the theoret-

ical predictions. First, which determinants drive confidentiality usage, and how much of the variation is explained by institutional versus commercial factors? Second, do firms differentially treat long-term and short-term confidentiality, as would be implied by the distinct strategic purposes identified in Section 3? Third, what role does appropriability play—and specifically, do incumbents and entrants differ in their adoption patterns, consistent with the predictions of the prior experience and value capture literatures?

For each period, we present product-level probit specifications that include only the variables of each driver category, as well as specifications with all variables jointly. All specifications include year fixed effects, and standard errors are clustered at the firm level.

$$P(Y_i = 1 \mid \mathbf{X}_i) = \Phi(\mathbf{X}_i'\boldsymbol{\beta})$$

Y_i is whether or not product i uses confidentiality, $\mathbf{X}_i$ is our vector of explanatory variables for product i, and Φ is the standard normal CDF. To improve interpretability, we also present the average marginal effects (AME) and marginal effect at the means (MEM) of the full probit specification. The AME calculates the marginal contribution of each variable to the probability of confidentiality usage at the observed values of the data, then averages over all data points. Alternatively, the MEM calculates the mean of each variable in the data, then evaluates the marginal contribution of each variable at those means. Since all variables are binary indicators, the AME and MEM for the kth variable in a dataset with N observations are:

$$\text{AME}_k = \frac{1}{N}\sum_{i=1}^{N} \Phi\left(\sum_{j\neq k} x_{ij}\beta_j + \beta_k\right) - \Phi\left(\sum_{j\neq k} x_{ij}\beta_j\right)$$

$$\text{MEM}_k = \Phi\left(\sum_{j\neq k} \beta_j \times \left[\frac{1}{N}\sum_{i=1}^{N} x_{ij}\right] + \beta_k\right) - \Phi\left(\sum_{j\neq k} \beta_j \times \left[\frac{1}{N}\sum_{i=1}^{N} x_{ij}\right]\right)$$

where i and j denote observations and variables, respectively.

In general, we prefer the AME since it corresponds to the sample average partial effect, as opposed to the partial effect of a "typical" product which may not actually exist in the data. Standard errors are calculated using the delta method:

$$\widehat{\text{Var}}_k = \mathbf{G}_k'\widehat{\boldsymbol{\Sigma}}\mathbf{G}_k$$

where $\widehat{\boldsymbol{\Sigma}}$ is the estimated variance-covariance matrix of $\widehat{\boldsymbol{\beta}}$ in the probit specification (with clustered standard errors) and $\mathbf{G}_k$ is the gradient of the AME_k or MEM_k with respect to $\boldsymbol{\beta}$.

6 Selection Equation Results

6.1 What are the main determinants (institutional vs. commercial) that explain confidentiality usage?

Table 8 presents the probit model for long-term confidentiality usage in the early period (2001–2007), which achieves a pseudo-R^2 of over 30% with all observables. Value capture variables alone (Column 3) explain 20% of the variation, suggesting that commercial considerations—rather than institutional factors such as regulatory familiarity or experience—are the primary driver of permanent secrecy adoption. This is a first indication that LTC operates primarily as an appropriability mechanism, consistent with the theoretical prediction that firms use permanent concealment to protect technically complex products from imitation.

Column (1) of Table 9 (and Figure 5) reports the AMEs for the baseline specification. The results align closely with the competitive pressures and value capture predictions outlined in Section 3. High technical complexity and the presence of Bluetooth and Wi-Fi significantly increase long-term confidentiality usage by 20% to 40%. This is consistent with the appropriability literature's emphasis on product design complexity as a means of protection against imitation: when the underlying innovation is difficult to codify, firms rely on secrecy to slow competitors' ability to reverse-engineer product features. Highly complex products such as phones increase LTC usage by over 30%, while simpler products—modules and transceivers, consumer products, and those outsourced to Asia—are associated with a 12% to 20% decrease in usage. Being in a competitive product category increases LTC usage by 15%, which is consistent with the prediction that firms facing intense rivalry have stronger incentives to withhold information (Jin 2005).

A notable finding concerns the interaction between confidentiality and traditional intellectual property. The effect of Bluetooth and Wi-Fi on LTC usage is entirely offset for Wi-Fi products with SEP protection, which exhibit a reduced usage probability of 30%. This substitution pattern is consistent with the theoretical prediction that patenting and secrecy serve as alternative appropriability mechanisms (Anton and Yao 2004). The substitution is specific to Wi-Fi, an emerging technology during this period in which SEPs conferred substantial market power.

Turning to locational variables, firm geography matters: Asian firms are 20% less likely to use LTC, whereas Silicon Valley firms have an increased probability of 12%. These patterns are consistent with the regulatory advantages prediction—domestic firms, particularly those in the innovation hub

most familiar with US regulatory processes, may be better positioned to navigate the confidentiality application process. However, variables capturing firm experience or regulatory complexity directly are either insignificant or small in magnitude. The absence of strong experience effects for LTC suggests that the barriers to adopting permanent confidentiality are low enough that even less experienced firms adopt it when the product-level incentives are present. This finding weighs against the strong form of capture theory, which would predict that established, regulatorily experienced firms should disproportionately exploit the LTC option.

Turning to short-term confidentiality, Tables 11 and 12 present the probit and marginal effects for the late period (2011–2021), respectively. The determinant structure differs sharply from that of LTC, both in explanatory power and in the identity of the key predictors. In general, less of short-term confidentiality usage is explainable (pseudo-R^2 of 20%), and coefficient magnitudes are smaller. Crucially, the composition of explanatory power reverses: whereas value capture variables dominate for LTC, firm experience variables (Column 2) account for most of the variation in STC usage, at 14%. This reversal is the first piece of evidence that short-term confidentiality serves a fundamentally different strategic purpose than its long-term counterpart.

Column (1) of Table 12 reveals that the predictors of STC usage align with the value capture and prior experience predictions rather than the competitive pressures that drive LTC. High regulatory complexity increases STC usage probability by 10%, consistent with the insurance interpretation: products that navigate more complex regulatory pathways face greater coordination risk between certification timing and commercial launch, making temporary protection more valuable. By contrast, technical complexity—the dominant predictor of LTC—is insignificant for STC. Neither the number of frequency ranges, nor the presence of Wi-Fi or Bluetooth, nor being a phone predicts short-term confidentiality usage. This result supports the theoretical distinction between appropriability-driven and timing-driven secrecy: the technical characteristics that create imitation risk are irrelevant to the decision to temporarily delay the public release of marketing materials.

The firm-level predictors of STC tell a story of established, brand-oriented firms managing the commercialization process. Outsourcing products to Asia decreases STC usage probability by 10%, while being an EU or North American firm increases it by 9% to 12%. Silicon Valley firms show an additional 10% increase. Firms with a large number of products have a 15% higher probability of using STC, consistent with the product portfolio literature's prediction that broad-portfolio firms face more complex coordination challenges across simultaneous product launches (Cabral 2000). Finally,

and in contrast to the substitution pattern observed for LTC, patenting is a *complement* to STC: products protected by SEPs (either Bluetooth or Wi-Fi) have a 12% higher probability of using short-term confidentiality. This complementarity is consistent with the view that firms deploying formal IP protection for their core technologies are also the firms with the most elaborate commercialization strategies—and hence the greatest need for temporary launch protection.

We perform a robustness check to account for TCBs (the ultimate form of institutional knowledge) in Appendix A2, and find no significant change in our results. Having established the main determinants of each form of confidentiality, we now turn to a direct comparison of the two.

6.2 How do the determinants differ for permanent (long-term) vs. commercialization (short-term) confidentiality usage?

For a more direct comparison, we examine the transition period (2008–2010) when both types of confidentiality coexist and can be modeled jointly. Table 10 presents a bivariate probit specification for both types of confidentiality, with long-term followed by short-term for each pair of columns. Figure 6 directly compares the marginal effects of LTC and STC during the transition period, offering the clearest view of how the same observable characteristics differentially predict the two forms of secrecy.

The comparison sharpens the theoretical distinction. First, regulatory and technical complexity operate in opposite directions for the two types of confidentiality. A product subject to three rule parts is 20% more likely to use STC and weakly *less* likely to use LTC. Conversely, products that are not unintentional radiators—those with at least one frequency range, indicating active transmission and greater technical sophistication—are 20% to 30% more likely to use LTC and slightly less likely to use STC. This is suggestive that regulatory complexity increases the coordination risk between certification and launch (favoring STC as insurance), while technical complexity increases the appropriability risk from information leakage (favoring LTC as trade secret protection).

Second, locational predictors are common to both forms of confidentiality. Being a domestic firm increases the probability of both LTC and STC usage by approximately 10%, though outsourcing to Asia is negatively predictive only for LTC usage (by 15%). This asymmetry suggests that the locational advantage in STC adoption reflects familiarity with the commercialization environment and branding norms of the US market, whereas the locational disadvantage for LTC could reflect that products assembled offshore tend to involve more commoditized designs with less to protect.

Third, the relationship between formal IP and confidentiality differs across the two types.

Unlike the univariate probit for the early period, Bluetooth devices protected by a SEP are 15% more likely to use LTC in the transition period, suggesting that the substitution pattern between patents and permanent secrecy may be technology- and period-specific. However, the substitution result for Wi-Fi SEPs persists: Wi-Fi products with SEP protection remain 10% less likely to use LTC. This specificity is consistent with the observation that Wi-Fi SEPs conferred particularly strong protection during its emergence as a dominant standard.

Overall, the bivariate probit confirms that the determinants of long-term and short-term confidentiality differ systematically, even when comparing them side by side within the same time period and institutional environment. Complex products in competitive markets gravitate toward permanent, IP-like secrecy—the domain of the appropriability literature. Products that adopt temporary protection instead emphasize insurance against regulatory delays during commercialization—a strategic concern rooted in the value capture and product portfolio literatures. The substitution between LTC and patenting, particularly for Wi-Fi technologies, echoes the broader finding in the appropriability literature that formal and informal IP mechanisms serve as alternatives (Levin et al. 1987, Cohen et al. 2000, Mezzanotti and Simcoe 2023). By contrast, the complementarity between STC and patenting suggests firms that invest in formal IP are also those with the most to lose from poorly timed product launches.

6.3 What role does appropriability play in short-term versus long-term confidentiality?

Having shown that regulatory capture plays no major role in explaining confidentiality usage, we turn to investigating appropriability. If LTC primarily serves an appropriability function, then its adoption should not differ systematically between incumbents and entrants once product-level characteristics are controlled for—any firm with a technically complex product has the same incentive to protect it. If STC instead serves a commercialization function, then incumbents—with their established brands, broader product portfolios, and greater launch coordination needs—should exhibit persistently higher adoption rates.

We define "incumbent" as a firm that existed before the relevant policy enactment: firms that filed for products before 2000 for long-term confidentiality, and before 2007 for short-term confidentiality. Figure 8 presents the raw adoption trends by year for incumbents and entrants. Two patterns stand out. For long-term confidentiality, incumbent and entrant usage does not differ significantly and converges over time. For short-term confidentiality, incumbent usage is persistently higher.

We confirm this asymmetry in Figure 9, which plots the predicted probabilities of confidential-

ity usage for incumbents versus entrants in a probit model, controlling for observables at their mean values. The specification includes all variables in Table 2, except those related to filing before 2000 and 2007 (to avoid collinearity), and augments the model with the incumbent indicator interacted with year fixed effects to capture dynamic differences. The results are consistent with the theoretical predictions. For long-term confidentiality, there are no significant differences between incumbents and entrants in the early period, and entrant usage quickly catches up in the later period. The convergence indicates that LTC adoption is driven primarily by product characteristics—technical complexity, competitive intensity, and the presence of imitable design features—rather than by firm-level advantages that only incumbents possess.

For short-term confidentiality, incumbents exhibit persistently higher usage than entrants, and the gap *increases* over time, even after controlling for all other observable characteristics. This widening divergence is consistent with the value capture and product portfolio literatures. Established firms with recognized brands face greater consequences from poorly timed product disclosures—premature revelation of a new model can cannibalize existing product sales, confuse retail partners, and undermine carefully sequenced marketing campaigns. Entrants, who typically lack established product lines and complex distribution channels, face fewer such coordination risks. The fact that the incumbent–entrant gap grows over time further suggests that the value of STC compounds with portfolio breadth and brand maturity, reinforcing its interpretation as commercialization insurance rather than a generic appropriability tool.

7 Event Study

The preceding analysis identifies the cross-sectional determinants of LTC and STC usage and demonstrates that these two forms of confidentiality serve distinct strategic purposes: LTC functions as a mechanism of appropriability for technically complex products, whereas STC serves as commercialization insurance for branded, experienced firms. A natural follow-up question is whether the *introduction* of STC had a measurable effect on firm behavior. If STC indeed functions as commercialization insurance, we would expect its availability to affect the *speed* at which products are brought to market, rather than the technical characteristics of those products. We focus on the behavioral effects of STC instead of LTC due to STC being enacted in the middle of our sample period, and the stable plateau it reached in take-up.

The arrival of STC in 2007 to 2008 applies universally to all firms and products. However, as documented in the descriptive patterns of Section 5, only approximately 30% of products select

into STC, and this share remains roughly constant over time. Despite this endogenous selection, we can observe that certain firm × product family combinations are ex-ante more prone to utilizing STC based on their time-invariant characteristics. We exploit this variation to construct a measure of differential *exposure* to the STC policy change and examine whether highly exposed firm–product cells exhibit different behavioral patterns after STC becomes available.

7.1 Identification Strategy

The identification relies on a three-step procedure to construct a continuous measure of ex-ante STC exposure at the firm × product family level. In the first step, we estimate a probit model of STC adoption using the 2011 to 2021 data, the post-adoption steady-state period that forms the basis of the main analysis in Section 5. For the purpose of this selection equation, we restrict the set of predictors to variables that are relatively time-invariant and capture firm-level and product-market-level characteristics: whether the firm filed before 2000, whether the firm filed for more than 20 products before 2007, firm location, whether the product market is an end market, whether the product is a module or transceiver, whether the product is a phone, and indicators for testing outsourcing. Importantly, this selection equation is used purely for prediction.

In the second step, we apply the estimated coefficients to the pre-period data (2001 to 2006) to generate predicted probabilities of STC usage for each product. We then average these predicted probabilities to the firm × product-family level, yielding a continuous exposure measure ranging from 0 to 1. In the data, actual exposure ranges from approximately 10% to 80% (Figure 10). The variable selection deliberately excludes outcome variables and time-varying product characteristics that are incomparable in the first and third time periods.

In the third step, we estimate an event study specification. We restrict the sample to firm × product-family cells that exist prior to 2007 and contain at least three observations in the 2001 to 2006 pre-period. The estimating equation takes the form:

$$Y_{fpt} = \sum_{k \neq -1} \beta_k \cdot \mathbf{1}[t - 2007 = k] \times \text{Exposure}_{fp} + \alpha_{fp} + \gamma_t + \varepsilon_{fpt} \tag{7.1}$$

where Y_{fpt} is the outcome for firm f in product family p at time t; Exposure_{fp} is the continuous predicted probability of STC usage constructed in the second step; α_{fp} are firm × product-family fixed effects; and γ_t are year fixed effects. The event time is normalized relative to 2007, with $k = -1$ as the omitted reference period. The coefficients β_k thus capture the differential change in outcomes for a one-unit increase in exposure (from 0% to 100%) at each event-time period, relative to 2006. Standard errors are clustered at the firm level.

All regressions are weighted by the inverse of the within-cell variance, defined as $\text{Var}(Y)/N_{fp}$, where N_{fp} is the number of product observations for firm f in product family p during the 2001 to 2006 pre-period. This weighting addresses noise arising from the fact that the exposure measure is an average across cells of varying size, so that cells with more underlying observations receive proportionally greater weight. The event study window covers 2001 to 2012, restricting the post-period to five years after implementation.

We examine three classes of outcomes. First, *product introduction speed*, measured as the median number of days between consecutive product applications within the cell. If STC serves as commercialization insurance, firms may file earlier for certification, reducing the gap between successive product introductions. Second, *complexity*, measured by the median number of frequency ranges (technical complexity) and the median number of rule parts (regulatory complexity) within the cell. Third, *product introductions*, measured as the log number of products in the cell per year. In addition, we conduct supplementary firm-level and product type level analyses using a fractional probit. At the firm-level, we measure product breadth, where the outcome is the number of product families or types in which the firm files applications, and exposure is constructed using only firm-level indicators. At the product type level, the outcome is number of survival-adjusted entrant firms, and exposure is constructed using a product type level probit. For a definition of survival-adjusted, refer to Section 4.5.

7.2 Results

Since the iPhone was released in 2007 and phones behave quite differently as a product category, we present all results both with the full sample and excluding phones. The use of medians at the cell level mitigates the influence of outliers, so the exclusion of phones does not substantially alter the main results.

Figure 11 presents the event study coefficients for the median number of days between consecutive product applications. The coefficients capture the differential change in time-to-market associated with a one-unit increase in STC exposure (0% to 100%), relative to the omitted year, 2006. The pre-period coefficients are approximately flat, consistent with the parallel trends assumption, and the post-period coefficients indicate a *decrease* in the median time between product introductions for firm $\times$ product-family cells with higher ex-ante STC exposure. The point estimates suggest that moving from 0% to 100% exposure is associated with a reduction of approximately 300 to 400 days (roughly one year) in the gap between successive product filings, relative to a pre-period mean of ap-

proximately 170 days. While this magnitude appears large, it reflects the full range of the continuous exposure measure. To benchmark the effect at empirically relevant values, assuming linearity, moving from the 25th percentile (20% exposure) to the 75th percentile (45% exposure) of the exposure distribution reduces the gap between successive product filings by approximately 90 days. The result is qualitatively unchanged when phones are excluded.

This finding is consistent with the interpretation of STC as commercialization insurance. Firms with a high ex-ante propensity to use STC—those that tend to be large, domestic, experienced—appear to bring products to market faster after STC becomes available. This aligns with the probit results from Sections 6, which show that STC adoption is driven by firm experience and branding considerations rather than technical complexity. If STC shields firms from the risk of premature disclosure during the period between certification and commercial launch, it may enable earlier filing by reducing the cost of a delay between regulatory approval and market readiness.

The remaining outcomes do not exhibit robust patterns. Median frequency ranges show an increase in the post-period when all products are included (with a mild pre-trend concern), but the effect is attenuated and borderline insignificant once phones are excluded (Figure 12). Similarly, the result for median rule parts exhibits a significant pre-trend when all products are included and a null effect once phones are excluded (Figure 13). The log number of products exhibits a significant pre-trend with or without phones, indicating that the same cells with high STC propensity were already increasing their product introductions prior to 2007. This result is therefore uninformative about the behavioral effect of STC (Figure 14).

At the firm level, the number of product families and product types in which a firm files applications shows a slight increase associated with higher exposure, but the effect is barely significant at the 95% confidence level with firm-clustered standard errors (Figure 15). The magnitudes are similar across both product-family and product-type measures, suggesting a consistent but imprecise pattern. At the product type level, there is a null effect on the number of survival adjusted entrants regardless of phone inclusion.

7.3 Interpretation

Several important caveats govern the interpretation of this exercise. First, the exposure measure is constructed from post-period adoption behavior (2011 to 2021 coefficients applied to pre-period characteristics). We are building a proxy for "latent susceptibility" to STC, relying on the assumption that, absent the introduction of STC, the outcomes of firm $\times$ product-family cells with different levels

of exposure would have followed a common trend. If firms that later experienced better outcomes are also those whose characteristics are predictive of STC adoption, the parallel trends assumption does not hold. We cannot entirely rule this out. Accordingly, this exercise should be understood as an exposure-based event study that identifies *associations* between predicted propensity and behavioral change, rather than as a treatment–control identification that yields causal estimates.

Second, the likely direction of bias is informative for interpreting our results. If anything, we expect the bias to be *positive*: ambitious, well-resourced firms that are predisposed to adopt STC may also independently reduce their time-to-market through other organizational or strategic means. This means that finding *zero* effects on non-speed outcomes is itself meaningful. If the exercise were simply picking up unobserved firm–product line quality, we would expect positive associations across all outcomes. The fact that the speed result stands alone, while other outcomes show no robust pattern, suggests that the time-to-market finding reflects more than pure endogeneity issues.

Overall, the event study exercise points to a single robust association: the propensity to use STC is associated with faster product introductions after the policy becomes available, but not with changes in technical complexity, regulatory complexity, the volume of products, product breadth, or firm entry. If the introduction of STC had any measurable effect on firm behavior, it was on the speed of execution—consistent with the view that firms could use STC to gain the option of an early release and to insure against delays arising from the coordination of certification timing with commercial launch. This finding reinforces the paper's central distinction: long-term confidentiality protects the *substance* of a product's technical design, functioning as a form of appropriability, whereas short-term confidentiality protects the *timing* of its commercial launch, functioning as commercialization insurance.

8 Conclusion

This study examines how firms strategically use confidentiality within the FCC's unlicensed spectrum certification system to manage intellectual property and commercialization risks. The study distinguishes between long-term confidentiality (LTC), which permanently conceals technical materials, and short-term confidentiality (STC), which temporarily protects marketing-related information. The evidence illustrates that these two forms of secrecy fulfill distinct economic and strategic purposes.

Our analysis reveals that LTC primarily operates as a mechanism of appropriability, being mostly adopted for technically complex products in competitive markets, and it functions as a sub-

stitute for formal intellectual property rights, such as patents. In contrast, STC serves mainly as a form of commercialization insurance, shielding firms from launch delays and coordination failures. Its use is concentrated among large, experienced, and brand-oriented firms rather than those producing technically complex devices. The incumbent–entrant analysis reinforces this distinction: LTC adoption converges across firm types, consistent with a product-level appropriability motive, whereas STC adoption remains persistently higher for incumbents, consistent with the commercialization needs of established firms managing complex product portfolios.

An exposure-based event study further supports the commercialization insurance interpretation. Firms with a high ex-ante propensity to use STC bring products to market faster after the policy becomes available, but show no differential changes in technical complexity, regulatory complexity, or the volume of products introduced. If STC had any measurable behavioral effect, it was on the speed of execution—enabling earlier certification filings by reducing the risk of premature disclosure during the interval between regulatory approval and commercial launch.

This divergence underscores that firms deploy secrecy not only to protect knowledge but also to manage timing, uncertainty, and visibility in their product markets. As unlicensed spectrum and private certification continue to expand, this study sheds light on the dual role of confidentiality—as both a protection of technological know-how and a buffer for commercialization dynamics.

References

Agarwal R, Ganco M, Ziedonis R (2009) Reputations for toughness in patent enforcement: implications for knowledge spillovers via inventor mobility. *Strategic Management Journal* 30(13):1349–1374.

American Certification Body, Inc (2018) *Information Regarding Different FCC Confidentiality Requests*. American Certification Body, Inc. (ACB), McLean, VA, URL https://acbcert.com/documents/misc-docs/Memo-Short-Term-Vs-Standard-Confidentiality.pdf, revised 04/05/18.

Anton JJ, Yao DA (2004) Little patents and big secrets: managing intellectual property. *RAND Journal of Economics* 1–22.

Arundel A (2001) The relative effectiveness of patents and secrecy for appropriation. *Research Policy* 30(4):611–624.

Baker S, Mezzetti C (2005) Disclosure as a strategy in the patent race. *The Journal of Law and Economics* 48(1):173–194.

Bertolotti F (2026) Patent term, innovation, and the role of technology disclosure externalities. *The Review of Economic Studies* rdag005.

Bhatia SS (1984) Ul's role in the changing world of telecommunications. *INTELEC '84 - International Telecommunications Energy Conference*, 208–213 (IEEE), URL http://dx.doi.org/10.1109/INTLEC.1984.4794109.

Bloom N, Schankerman MS, Van Reenen J (2013) Identifying technology spillovers and product market rivalry. *Econometrica* 81(4):1347–1393.

Board O (2009) Competition and disclosure. *The Journal of Industrial Economics* 57(1):197–213.

Bunch DS, Smiley R (1992) Who deters entry? evidence on the use of strategic entry deterrents. *The Review of Economics and Statistics* 509–521.

Bushee BJ, Keusch T, Kim-Gina J (2024) Co-opetition and the firm's information environment. *Management Science* 70(10):6917–6941.

Cabral LMB (2000) Stretching firm and brand reputation. *RAND Journal of Economics* 658–673.

Carpenter D, Moss DA, eds. (2013) *Preventing Regulatory Capture: Special Interest Influence and How to Limit it* (Cambridge: Cambridge University Press).

Carpenter DP (2002) Groups, the media, agency waiting costs, and FDA drug approval. *American Journal of Political Science* 46(3):490–505.

Carpenter DP (2004) Protection without capture: product approval by a politically responsive, learning regulator. *American Political Science Review* 98(4):613–631.

Carpenter DP, Moffitt SI, Moore CD, Rynbrandt RT, Ting MM, Yohai I, Zucker EJ (2010) Early entrant protection in approval regulation: Theory and evidence from FDA drug review. *The Journal of Law, Economics, & Organization* 26(3):515–545.

Cassiman B, Veugelers R (2002) R&d cooperation and spillovers: some empirical evidence from belgium. *American Economic Review* 92(4):1169–1184.

Ceccagnoli M (2009) Appropriability, preemption, and firm performance. *Strategic Management Journal* 30(1):81–98.

Chandra A, Kao J, Miller KL, Stern AD (2026) Regulatory incentives for innovation: The fda's breakthrough therapy designation. *The Review of Economics and Statistics* 108(2):470–484.

Chu Y, Dasgupta S, Ma F (2025) Competition, cannibalization, and new product introductions: Evidence from the pharmaceutical industry, mimeo.

Cockburn IM, MacGarvie MJ (2011) Entry and patenting in the software industry. *Management Science* 57(5):915–933.

Cohen WM, Nelson RR, Walsh JP (2000) Protecting their intellectual assets: Appropriability conditions and why US manufacturing firms patent (or not). Working Paper w7552, National Bureau of Economic Research.

Conti A, Ortega L, Sung E (2025) Sharpen your sword for litigation: incumbent strategic reaction to the threat of entry. *Management Science* 71(1):803–824.

Cookson JA (2018) Anticipated entry and entry deterrence: Evidence from the american casino industry. *Management Science* 64(5):2325–2344.

Crass D, Garcia-Valero F, Pitton F, Rammer C (2019) Protecting innovation through patents and trade secrets: Evidence for firms with a single innovation. *International Journal of the Economics of Business* 26(1):117–156.

Cunningham C, Kapacinskaite A (2026) Keeping invention confidential. *Management Science* 72(2):1658–1679.

De Figueiredo JM, Kyle MK (2006) Surviving the gales of creative destruction: The determinants of product turnover. *Strategic Management Journal* 27(3):241–264.

de Rassenfosse G, Zhou L (2020) Patents and supra-competitive prices: Evidence from consumer products, available at SSRN 3756359.

Dye RA, Sridhar SS (1995) Industry-wide disclosure dynamics. *Journal of Accounting Research* 33(1):157–174.

Ethiraj SK, Levinthal D, Roy RR (2008) The dual role of modularity: Innovation and imitation. *Management Science* 54(5):939–955.

Fang TP, Greenstein S (2025) Where the cloud rests: The economic geography of data centers. *Strategy Science* 10(4):404–420.

Farronato C, Zervas G (2022) Consumer reviews and regulation: Evidence from nyc restaurants. Working Paper w29715, National Bureau of Economic Research.

Federal Communications Commission (2004) OET equipment authorization system upgrade permits electronic submittal of short-term confidentiality requests. Washington, D.C., URL https://docs.fcc.gov/public/attachments/DA-04-1705A1.pdf, public Notice DA 04-1705.

Federal Communications Commission (2011) Confidentiality procedures detail. Draft Laboratory Division Publications Report KDB 726920 D02 v01, Office of Engineering and Technology, Laboratory Division, draft for Review.

Fontana R, Greenstein S (2021) Platform leadership and supply chains: Intel, centrino, and the restructuring of wi-fi supply. *Journal of Economics & Management Strategy* 30(2):259–286.

Furman JL, Nagler M, Watzinger M (2021) Disclosure and subsequent innovation: Evidence from the patent depository library program. *American Economic Journal: Economic Policy* 13(4):239–270.

Ganco M, Ziedonis RH, Agarwal R (2015) More stars stay, but the brightest ones still leave: Job hopping in the shadow of patent enforcement. *Strategic Management Journal* 36(5):659–685.

Gill D (2008) Strategic disclosure of intermediate research results. *Journal of Economics & Management Strategy* 17(3):733–758.

Graham SJH (2004) Hiding in the patent's shadow: Firms' uses of secrecy to capture value from new discoveries. Technical report, GaTech TI:GER Working Paper Series.

Greenstein N, Fontana R, Greenstein S, Zhang R, Zhang H, Olivarez O, Kim DY (2026) Wireless communications equipment markets: Evolution, classification, and measurement, unpublished manuscript.

Gross DP (2023) The hidden costs of securing innovation: the manifold impacts of compulsory invention secrecy. *Management Science* 69(4):2318–2338.

Grossman SJ (1981) The informational role of warranties and private disclosure about product quality. *The Journal of Law and Economics* 24(3):461–483.

Grossman SJ, Hart OD (1980) Disclosure laws and takeover bids. *Journal of Finance* 35:323–334.

Hall B, Helmers C, Rogers M, Sena V (2014) The choice between formal and informal intellectual property: a review. *Journal of Economic Literature* 52(2):375–423.

Hall BH, Helmers C (2024) *The Economics of Innovation and Intellectual Property* (Oxford University Press).

Hazlett TW (2017) *The Political Spectrum: The Tumultuous Liberation of Wireless Technology, from Herbert Hoover to the Smartphone* (New Haven, CT: Yale University Press).

Hegde D, Herkenhoff K, Zhu C (2023) Patent publication and innovation. *Journal of Political Economy* 131(7):1845–1903.

Hegde D, Luo H (2018) Patent publication and the market for ideas. *Management Science* 64(2):652–672.

Hou Y, Png IP, Xiong X (2023) When stronger patent law reduces patenting: Empirical evidence. *Strategic Management Journal* 44(4):977–1012.

Jansen J (2010) Strategic information disclosure and competition for an imperfectly protected innovation. *The Journal of Industrial Economics* 58(2):349–372.

Jin GZ (2005) Competition and disclosure incentives: An empirical study of HMO's. *RAND Journal of Economics* 36:93–113.

Kao J (2025) Information disclosure and competitive dynamics: Evidence from the pharmaceutical industry. *Management Science* 71(7):5948–5970.

Kim DY, Fontana R, Greenstein S (2024) Open devices and slices: Evidence from wi-fi equipment. Working Paper 24-045, Harvard Business School Technology & Operations Mgt. Unit.

Knapp J, Wall A (1997) Considerations for streamlining the fcc equipment authorization program. *IEEE 1997, EMC, Austin Style. IEEE 1997 International Symposium on Electromagnetic Compatibility. Symposium Record (Cat. No. 97CH36113)*, 23–28 (Austin, TX, USA: IEEE), URL http://dx.doi.org/10.1109/ISEMC.1997.667533.

Koo B, Jangho Y, Wright BD (2025) Protecting intermediate innovations when ideas are scarce: Patents or secrecy? *Journal of Economics & Management Strategy* .

Kyle MK (2002) Entry in pharmaceutical markets, mimeo, Program on the Pharmaceutical Industry, Sloan School of Management, Massachusetts Institute of Technology.

Levin RC, Klevorick AK, Nelson RR, Winter SG, Gilbert R, Griliches Z (1987) Appropriating the returns from industrial research and development. *Brookings Papers on Economic Activity* 1987(3):783–831.

Markopoulos PM, Hosanagar K (2018) A model of product design and information disclosure investments. *Management Science* 64(2):739–759.

Marx M, Strumsky D, Fleming L (2009) Mobility, skills, and the michigan non-compete experiment. *Management Science* 55(6):875–889.

Mezzanotti F, Simcoe T (2023) Innovation and appropriability: Revisiting the role of intellectual property. Working Paper w31428, National Bureau of Economic Research.

Milgrom PR (1981) Good news and bad news: Representation theorems and applications. *The Bell Journal of Economics* 380–391.

Milgrom PR, Levin J, Eilat A (2011) The case for unlicensed spectrum, available at SSRN 1948257.

Mohammadi A, Khashabi P (2017) Embracing the sharks: The impact of information disclosure on CVC investments. *Academy of Management Proceedings*, volume 2017, 15980.

Moorthy KS, Png IPL (1992) Market segmentation, cannibalization, and the timing of product introductions. *Management Science* 38(3):345–359.

Okuno-Fujiwara M, Postlewaite A, Suzumura K (1990) Strategic information revelation. *The Review of Economic Studies* 57(1):25–47.

Olson M (1997) Firm characteristics and the speed of FDA approval. *Journal of Economics & Management Strategy* 6(2):377–401.

Peltzman S (1976) Towards a more general theory of regulation. *Journal of Law and Economics* 19:211–240.

Pepall LM, Richards DJ (2002) The simple economics of brand stretching. *The Journal of Business* 75(3):535–552.

Png IPL (2017a) Law and innovation: Evidence from state trade secrets laws. *Review of Economics*

and Statistics 99(1):167–179.

Png IPL (2017b) Secrecy and patents: Theory and evidence from the uniform trade secrets act. *Strategy Science* 2(3):176–193.

Ponce CJ (2007) More secrecy... more knowledge disclosure? on disclosure outside of patents. Working Paper 07-72, Departamento de Economia, Universidad Carlos III de Madrid.

Rogers P (2023) Regulating the innovators: approval costs and innovation in medical technologies. Technical report, Cato Institute.

Schneider C (2008) Fences and competition in patent races. *International Journal of Industrial Organization* 26(6):1348–1364.

Sim K (2021) Optimal use of patents and trade secrets for complex innovations. *International Journal of Industrial Organization* 79:102788.

Starr EP, Prescott JJ, Bishara ND (2021) Noncompete agreements in the US labor force. *The Journal of Law and Economics* 64(1):53–84.

Stern AD (2017) Innovation under regulatory uncertainty: Evidence from medical technology. *Journal of Public Economics* 145:181–200.

Stigler GJ (1971) The theory of economic regulation. *The Bell Journal of Economics and Management Science* 2(1):3–21.

Stigler GJ (1975) *The Citizen and the State: Essays on Regulation* (Chicago: University of Chicago Press).

Stivers AE (2004) Unraveling of information: Competition and uncertainty. *The BE Journal of Theoretical Economics* 4(1).

Trlifaj 2025) Government transparency and innovation: Evidence from wireless products. URL https://arxiv.org/abs/2510.19377.

Viscusi WK (1992) *Fatal Tradeoffs* (New York: Oxford University Press).

Viscusi WK, Harrington JE, Sappington DEM (2018) *Economics of Regulation and Antitrust* (Cambridge, MA: The MIT Press), 5 edition.

Zaby AK (2010) Losing the lead: The patenting decision in the light of the disclosure requirement. *Economics of Innovation and New Technology* 19(2):147–164.

Zhang H, Greenstein S, Fontana R, Kim DY (2026) Regulatory reform and market evolution in the spectrum commons, working Paper.

Zhang T (2012) Patenting in the shadow of independent discoveries by rivals. *International Journal of Industrial Organization* 30(1):41–49.

Ziedonis RH (2004) Don't fence me in: Fragmented markets for technology and the patent acquisition strategies of firms. *Management Science* 50(6):804–820.

Figures

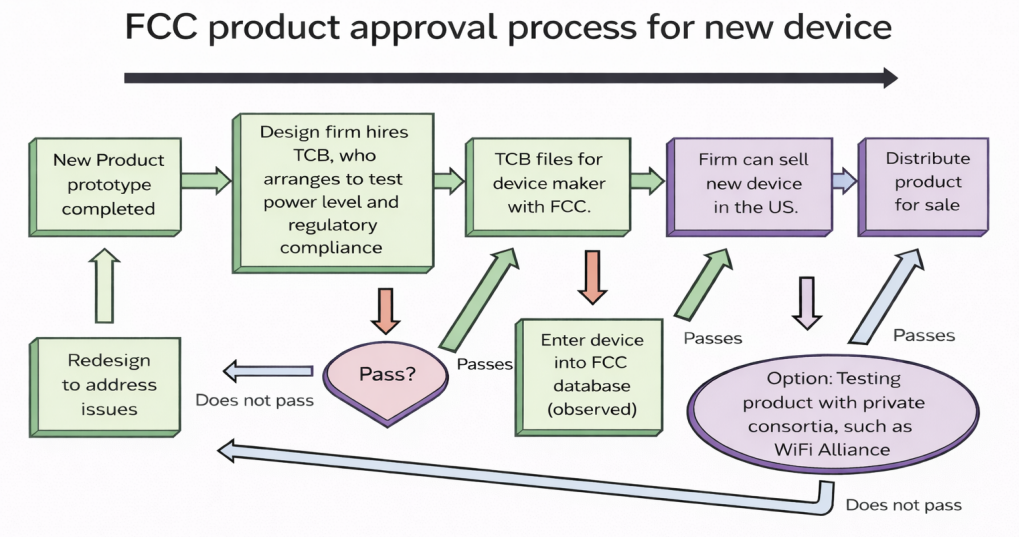


Figure 1: Current-day FCC Approval Process. This figure presents the current-day approval process for a new device using unlicensed spectrum. Once a device prototype is complete, the manufacturer hires a Telecommunication Certification Body (TCB) to test and subsequently file paperwork with the FCC. After the TCB applies on behalf of the manufacturer, we observe the device and application materials in the FCC database. Once the FCC approves of the device, the manufacturer can begin official marketing and sale of the device in the United States.

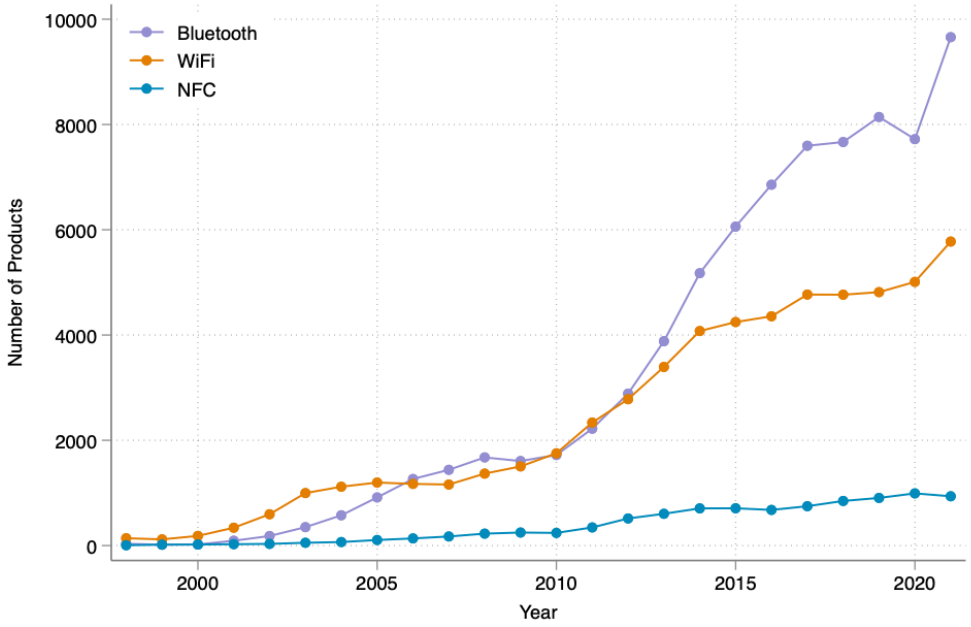


Figure 2: Number of Products Using Bluetooth, WiFi and Near-Field Communication. This figures plots the number of products which use Bluetooth, Near-Field Communication (NFC), and/or WiFi (any GHz) from 1998 to 2021. Usage is determined from the tested frequency bands. Products may simultaneously fall under multiple standards.

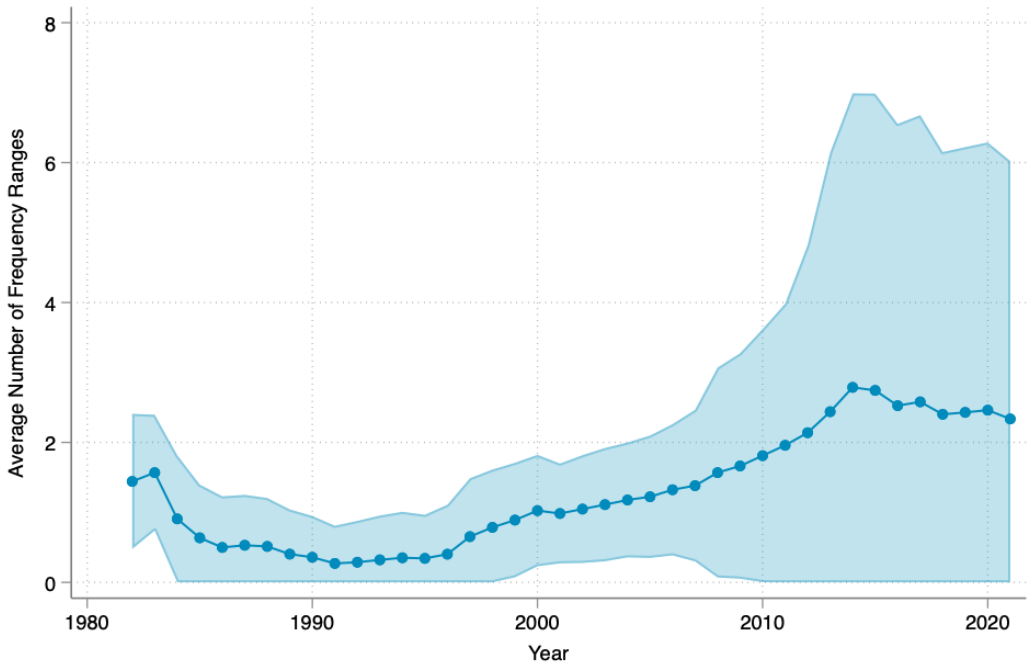


Figure 3: Average Number of Frequency Ranges (± 1 SD). This figure plots the average number of frequency ranges per product as a connected dot line from 1982 to 2021, along with ± one standard deviation in the shaded area. The lowest number of frequency ranges is zero, corresponding to unintentional radiators.

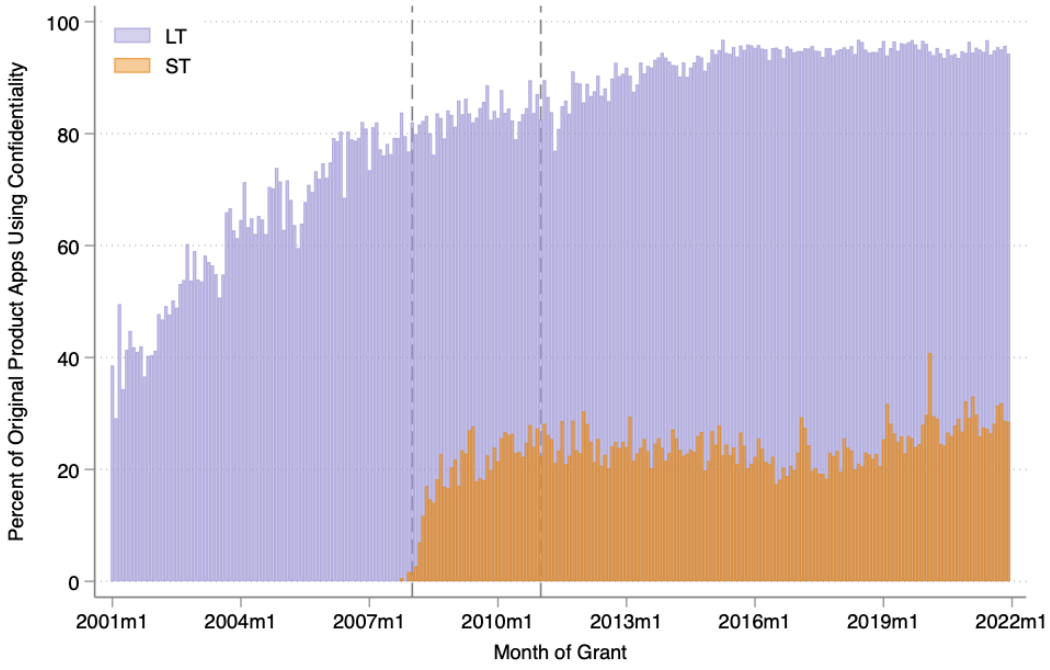


Figure 4: Percent of Products Using Long-Term and Short-Term Confidentiality. This figure plots the monthly raw percent of products using long-term (LT) and short-term (ST) confidentiality in purple and orange, respectively, from 2001 through 2021. The dashed lines separate the data sample into our three main periods of analysis: the early period (2001 to 2007), the transition period (2007 to 2010), and the late period (2011 to 2021).

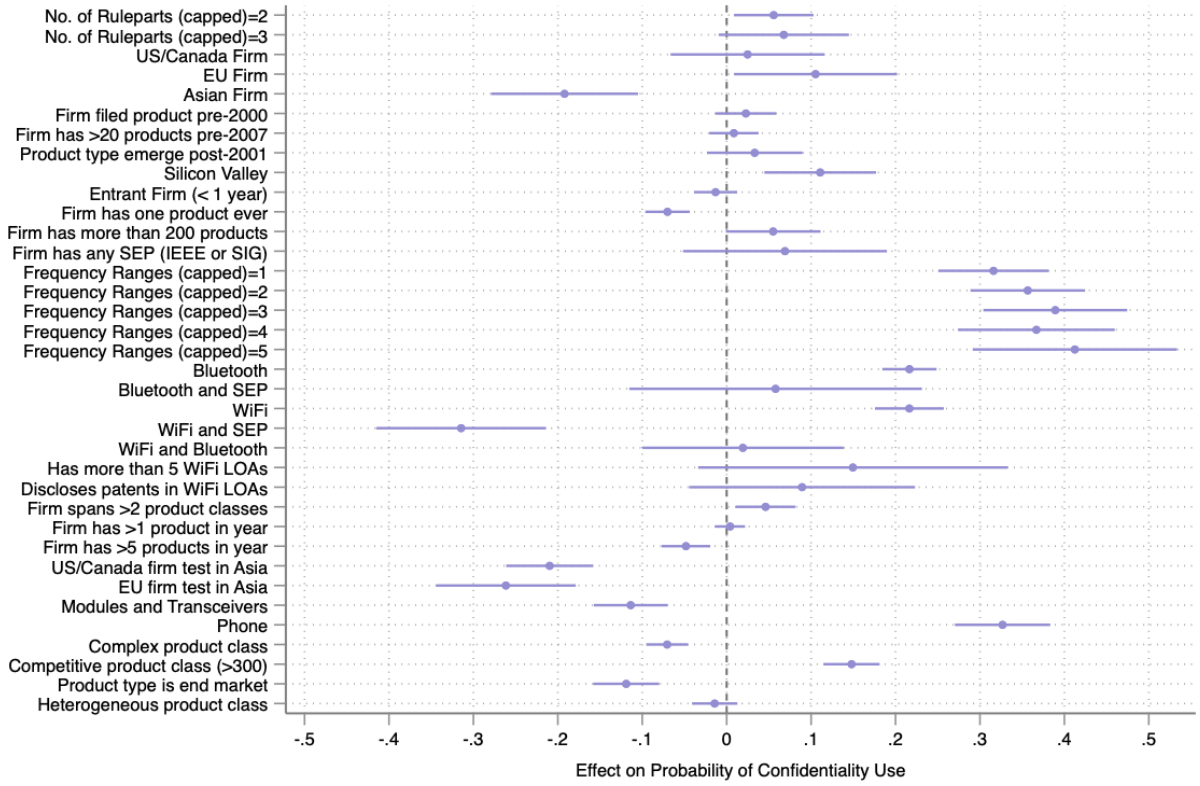


Figure 5: Marginal Effects of Product-Level LT Confidentiality from 2001 to 2007. This figure plots the effect of each explanatory variable on the probability of long-term confidentiality use with data from 2001 to 2007. We first run a probit regression of confidentiality usage on explanatory variables, with standard errors clustered at the firm level. Using the probit coefficients, we compute the marginal contribution of each variable to the probability of confidentiality usage at the observed values of the data, then average over all data points. Standard errors are computed with the delta method.

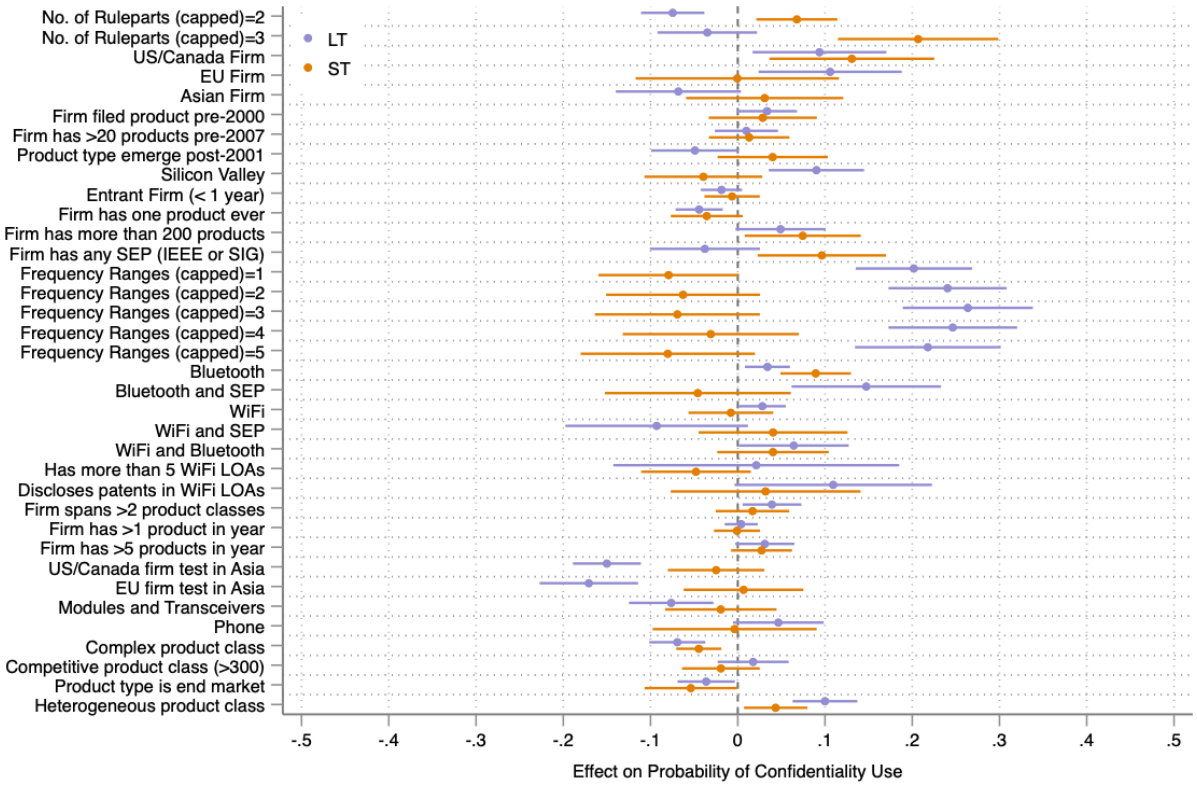


Figure 6: Marginal Effects of Product-Level LT and ST Confidentiality from 2008 to 2010. This figure plots the effect of each explanatory variable on the probability of long-term (LT) and short-term (ST) confidentiality use in purple and orange, respectively. We use data from 2008 to 2010. We first run a bivariate probit regression of confidentiality usage on explanatory variables, with standard errors clustered at the firm level. Using the bivariate probit coefficients, we compute the marginal contribution of each variable to the probability of confidentiality usage at the observed values of the data, then average over all data points. Standard errors are computed with the delta method.

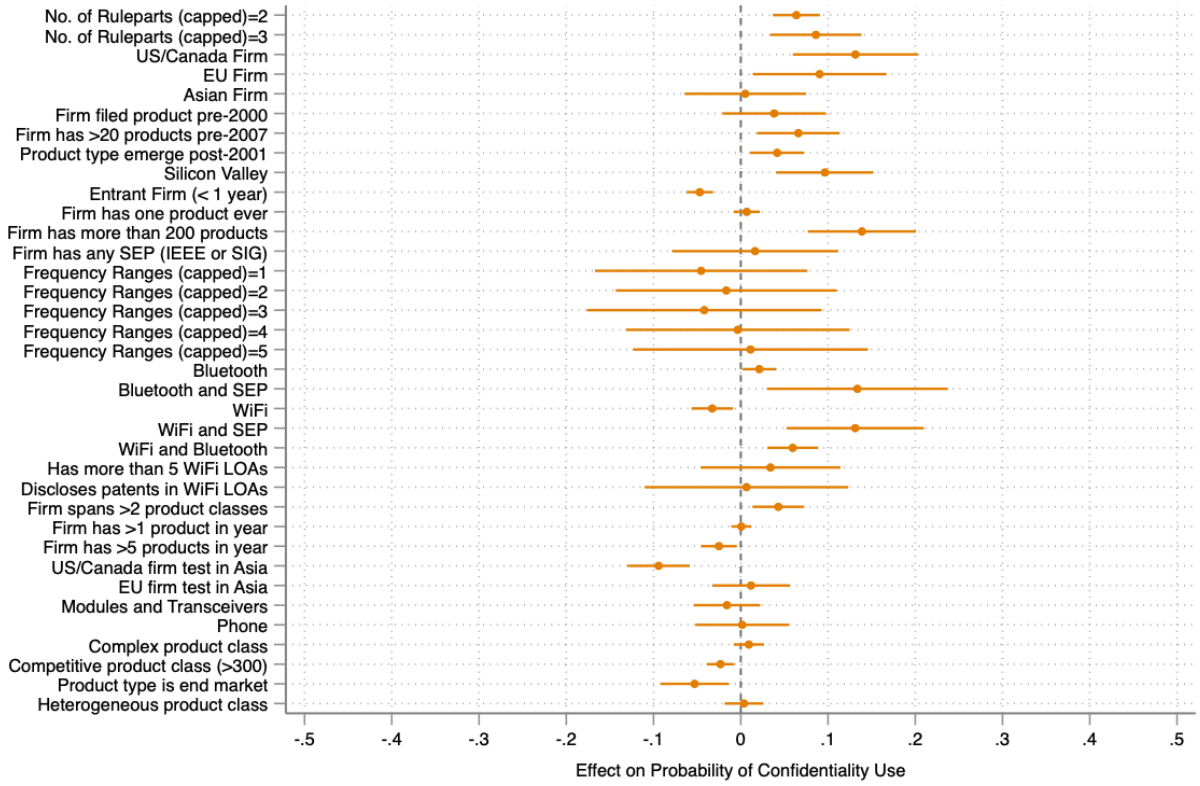


Figure 7: Marginal Effects of Product-Level ST Confidentiality from 2011 Onwards. This figure plots the effect of each explanatory variable on the probability of short-term confidentiality use with data from 2011 to 2021. We first run a probit regression of confidentiality usage on explanatory variables, with standard errors clustered at the firm level. Using the probit coefficients, we compute the marginal contribution of each variable to the probability of confidentiality usage at the observed values of the data, then average over all data points. Standard errors are computed with the delta method.

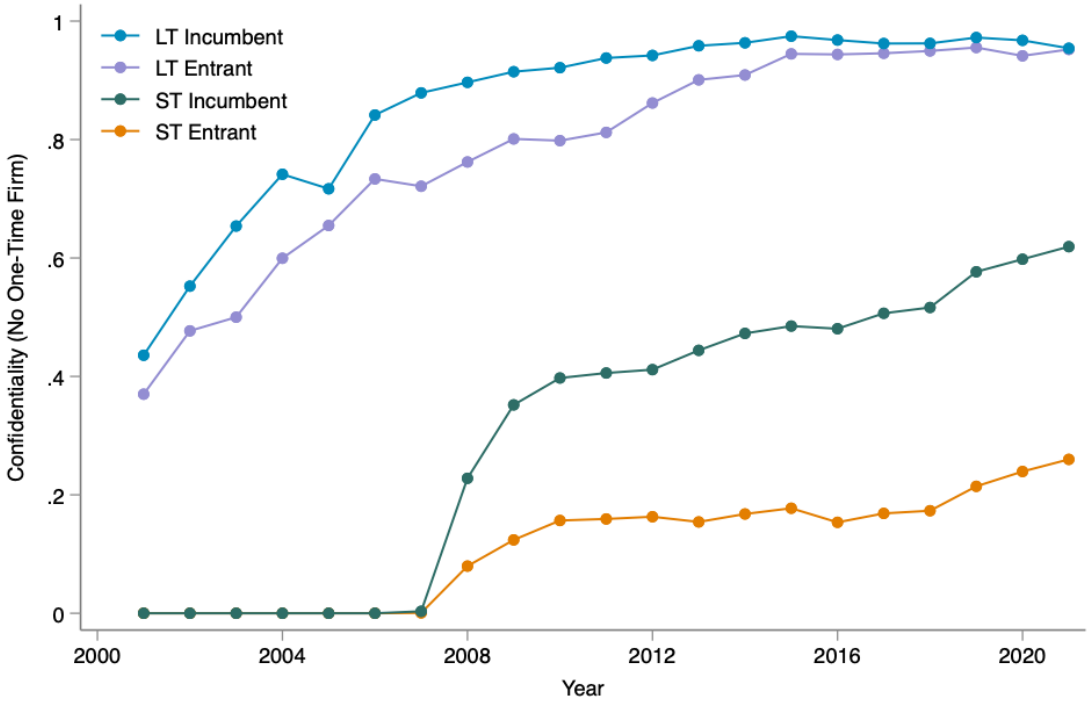
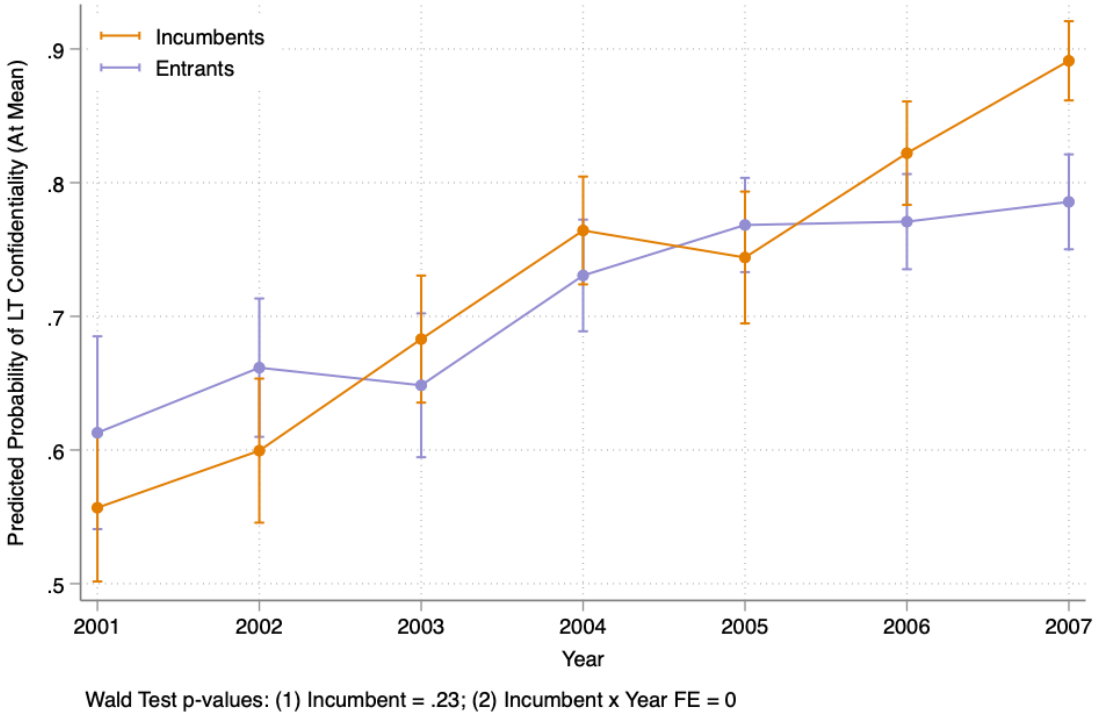
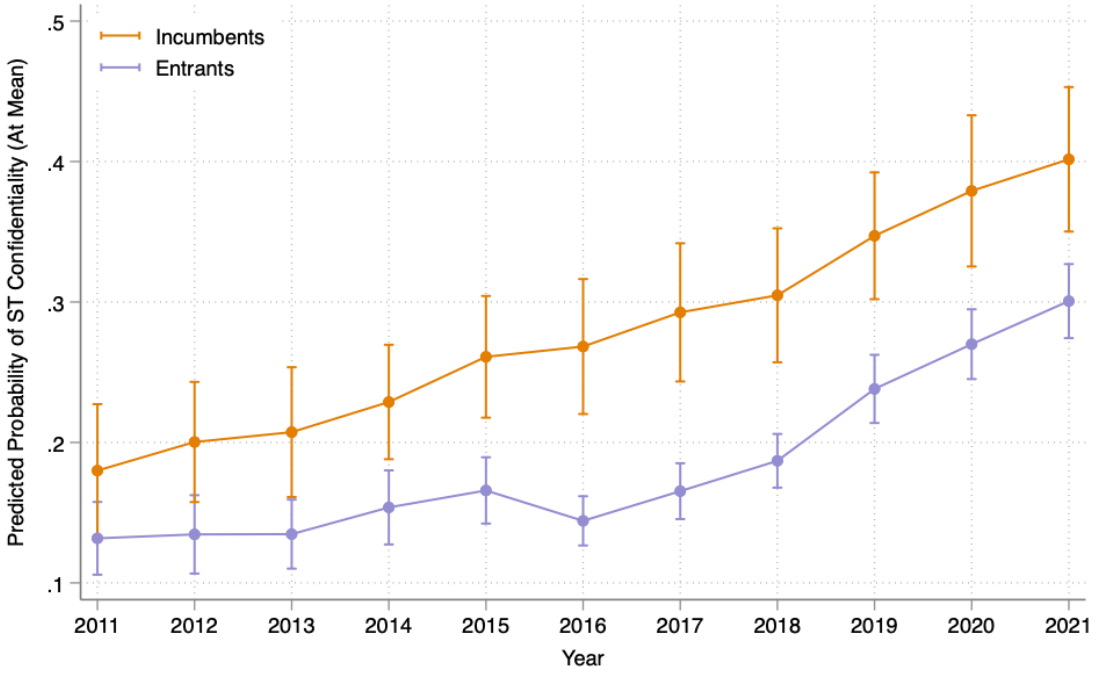


Figure 8: Percent of Confidentiality Usage (Excluding One-Time Firms). This figure plots the raw percentage of products from incumbent and entrant firms using long-term (LT) and short-term (ST) confidentiality from 2001 to 2021, excluding products from firms that appear only once in the database. We define incumbent as a firm that existed prior to policy enactment. Thus, incumbents are firms that filed products with the FCC prior to 2000 and 2007 for long-term and short-term confidentiality, respectively.



Wald Test p-values: (1) Incumbent = .23; (2) Incumbent x Year FE = 0



Wald Test p-values: (1) Incumbent = .08; (2) Incumbent x Year FE = .31

Figure 9: Predicted Probability of Confidentiality Usage. This figure plots the probability of using long-term (top panel) and short-term (bottom panel) confidentiality for incumbents (orange) and entrants (purple), holding all other variables constant at their mean values. For each panel, we first run a probit regression of confidentiality usage on explanatory variables, then use the probit coefficients to calculate predicted probabilities. We perform two hypotheses tests: (1) the coefficient on being an incumbent is significant and (2) the coefficients of being an incumbent interacted with year fixed effects are jointly significant. We present the p-values of the Wald tests below each panel.

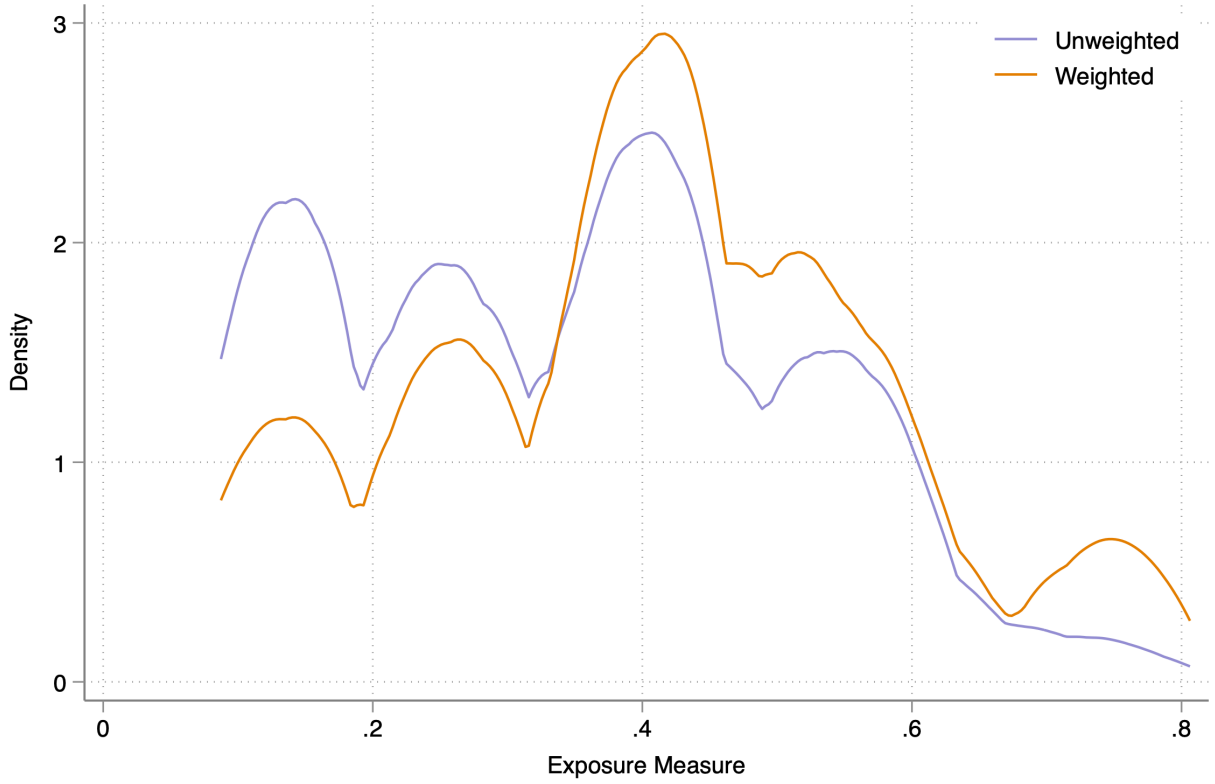


Figure 10: Kernel Density of Predicted STC Exposure. This figure plots the kernel density of the predicted probability of STC usage at the firm × product-family level, constructed by applying coefficients from a probit selection equation estimated on 2011 to 2021 data to pre-period (2001 to 2006) characteristics. The purple line shows the unweighted density; the orange line shows the density weighted by the inverse variance of cell size. The support of the exposure measure ranges from approximately 10% to 80%.

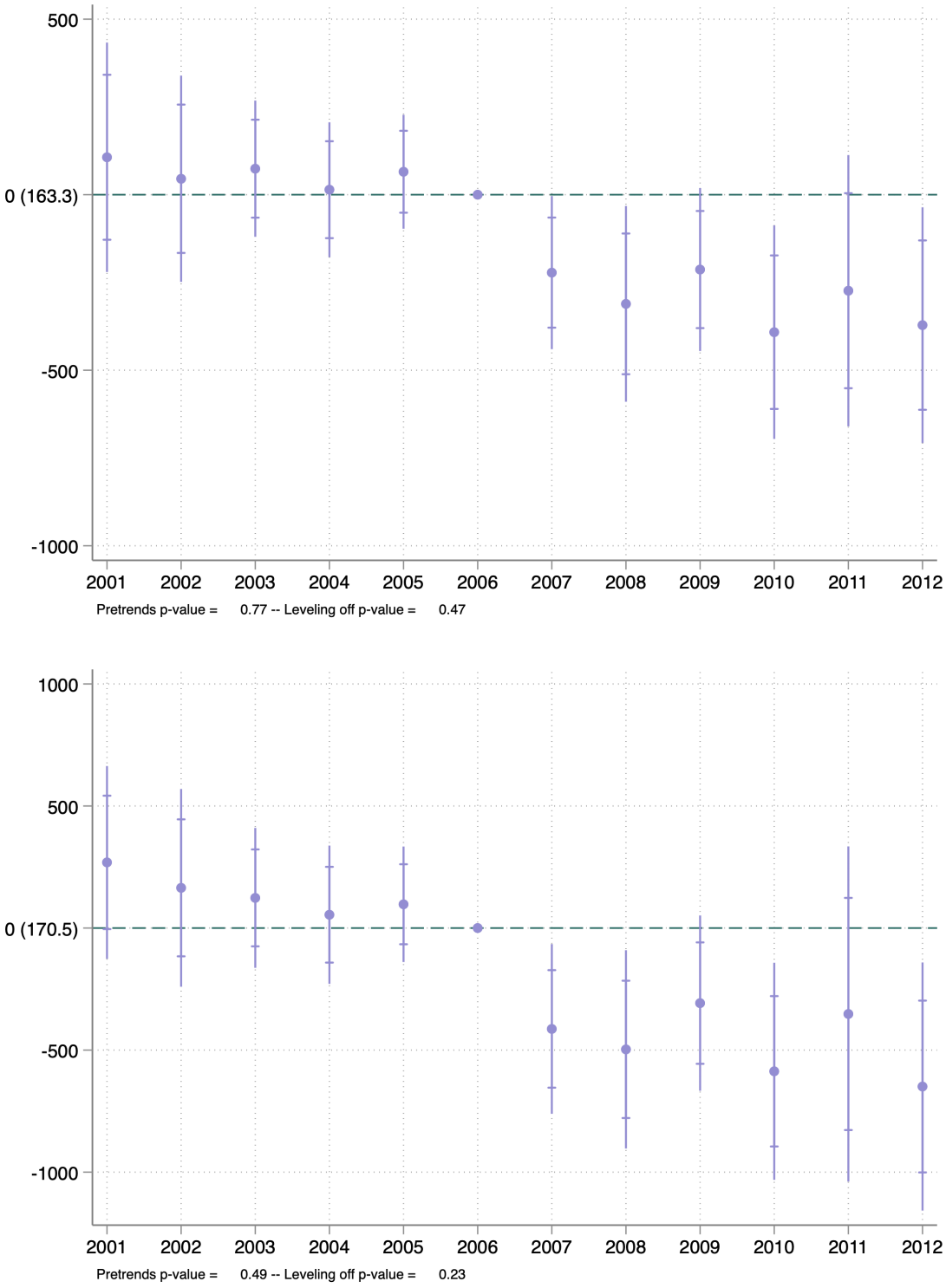


Figure 11: STC Event Study: Median Days Between Product Applications. This figure plots the event study coefficients relative to 2006, where the outcome is the median number of days between consecutive product applications within a firm × product-family cell. The top panel includes all products; the bottom panel excludes phones. All regressions are weighted by the inverse variance of cell size. Standard errors are clustered at the firm level. We plot the 95% confidence intervals.

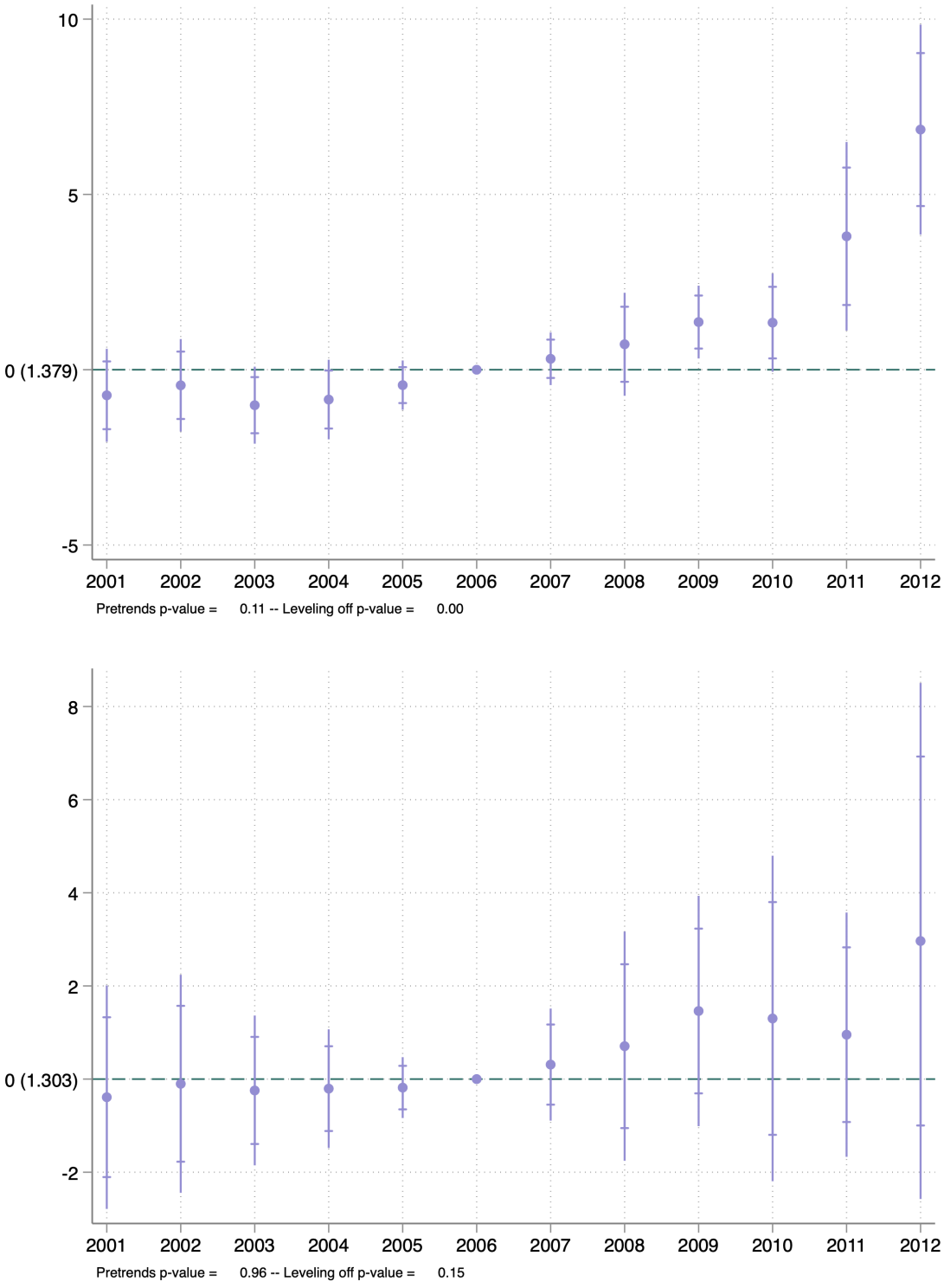


Figure 12: STC Event Study: Median Number of Frequency Ranges. This figure plots the event study coefficients relative to 2006, where the outcome is the median number of frequency ranges within a firm × product-family cell. The top panel includes all products; the bottom panel excludes phones. All regressions are weighted by the inverse variance of cell size. Standard errors are clustered at the firm level. We plot the 95% confidence intervals.

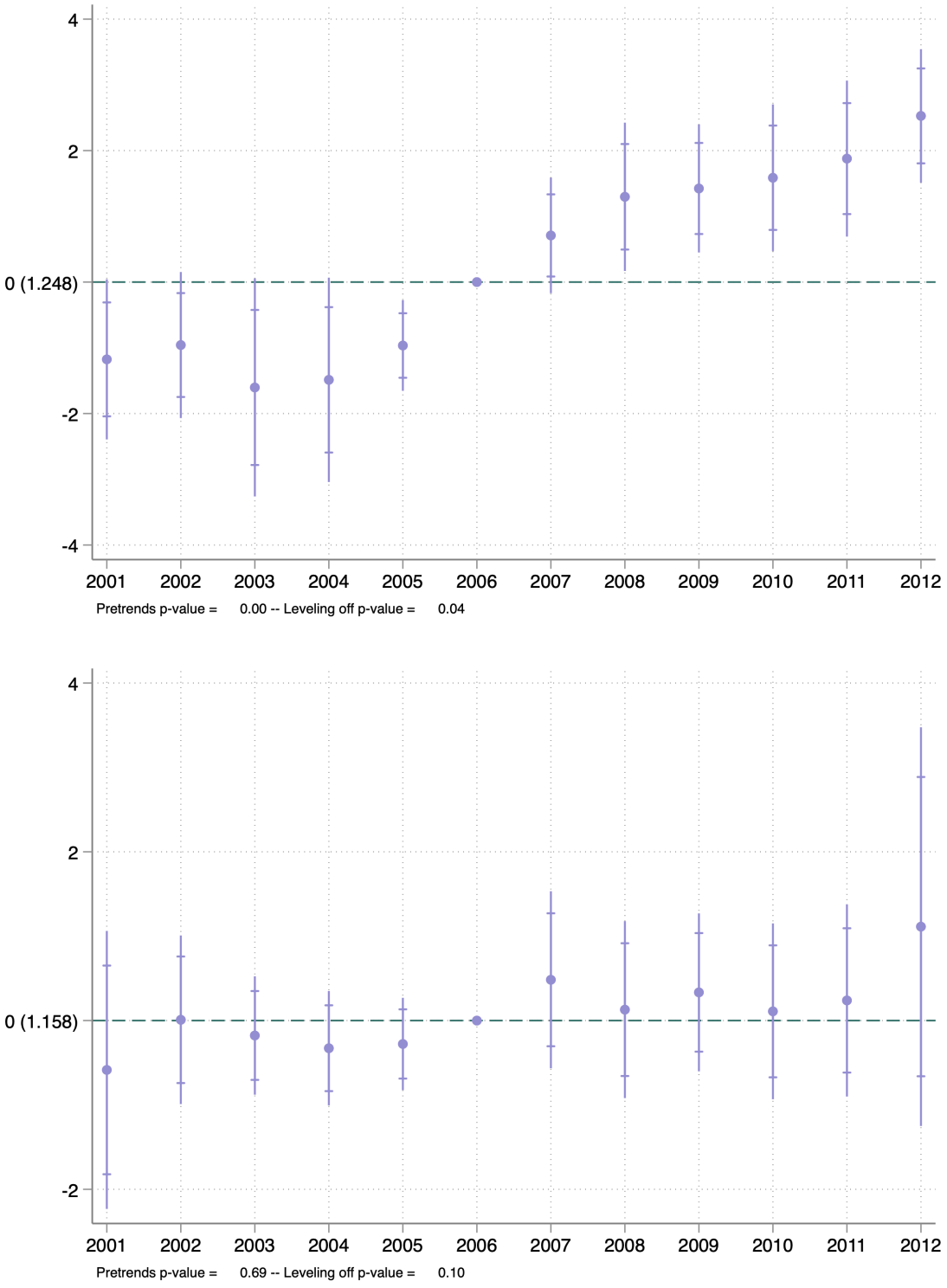


Figure 13: STC Event Study: Median Number of Rule Parts. This figure plots the event study coefficients relative to 2006, where the outcome is the median number of rule parts within a firm × product-family cell. The top panel includes all products; the bottom panel excludes phones. All regressions are weighted by the inverse variance of cell size. Standard errors are clustered at the firm level. We plot the 95% confidence intervals.

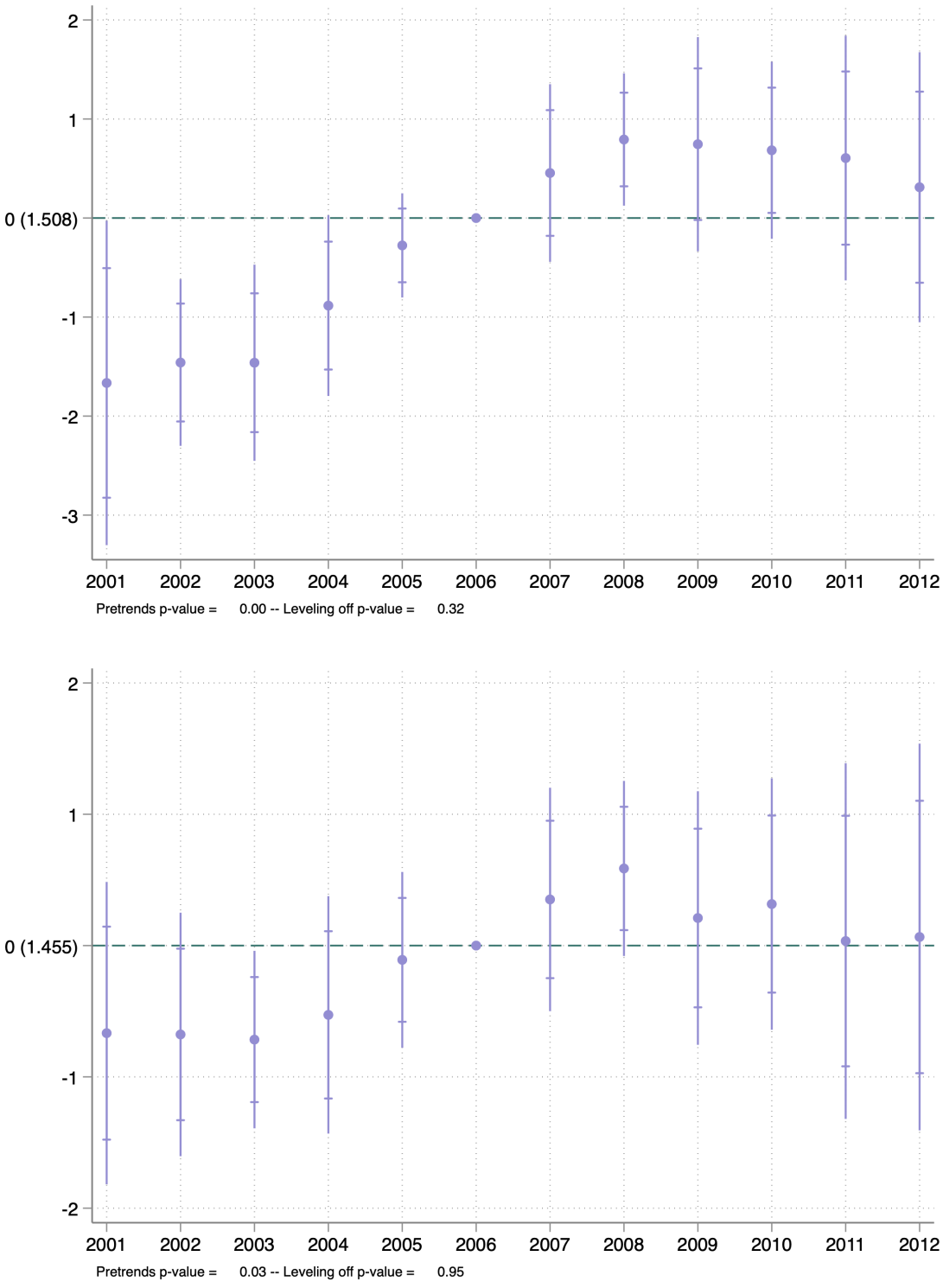


Figure 14: STC Event Study: Log Number of Products. This figure plots the event study coefficients relative to 2006, where the outcome is the log number of products for a firm × product-family cell. The top panel includes all products; the bottom panel excludes phones. All regressions are weighted by the inverse variance of cell size. Standard errors are clustered at the firm level. We plot the 95% confidence intervals.

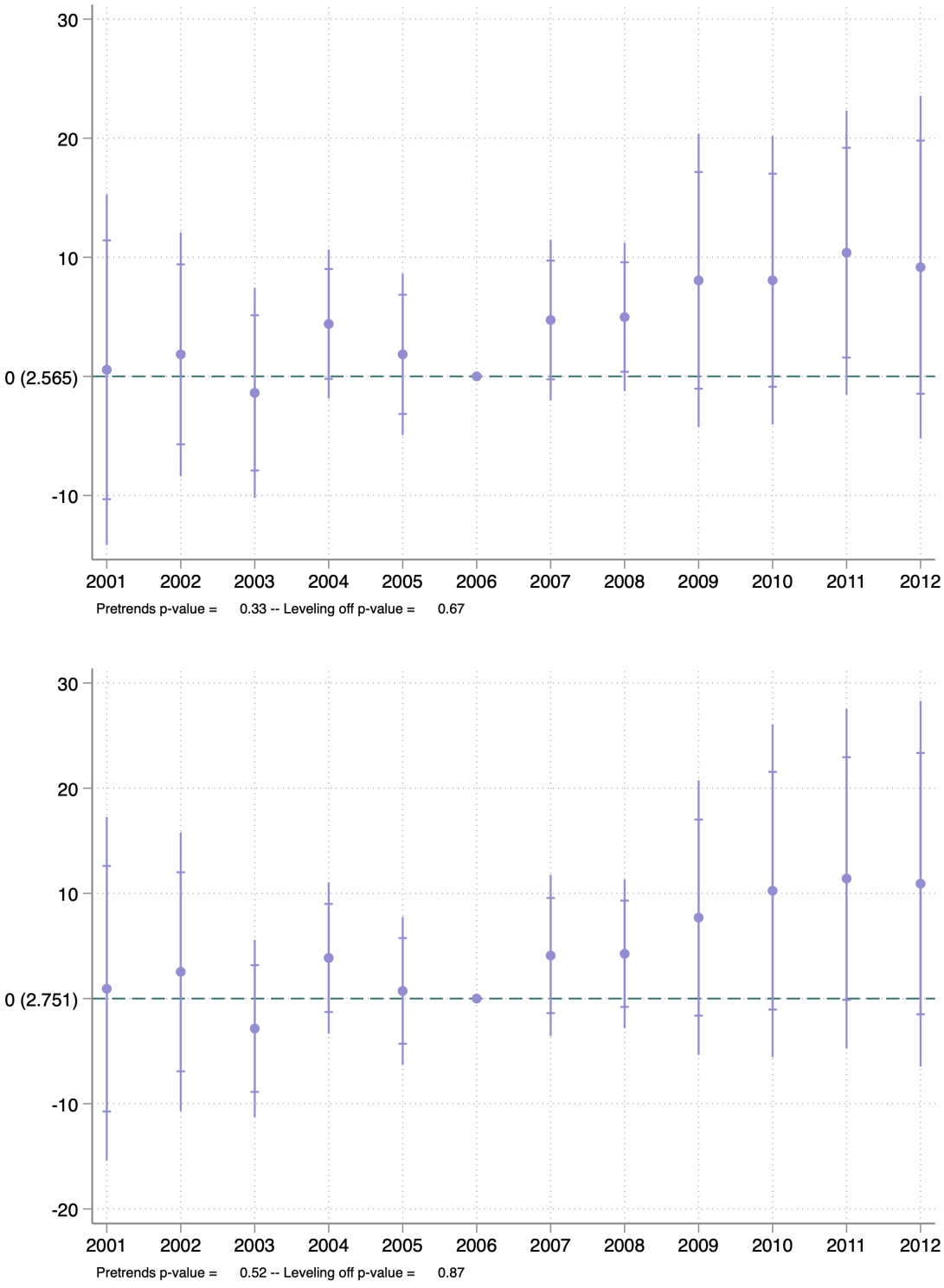


Figure 15: Firm-Level STC Event Study: Product Breadth. This figure plots the event study coefficients relative to 2006 from a firm-level specification, where the outcome is the number of product families (top panel) or product types (bottom panel) in which a firm has granted products. Exposure is constructed using firm-level indicators only. All regressions are weighted by the inverse variance of the number of products. Standard errors are clustered at the firm level.

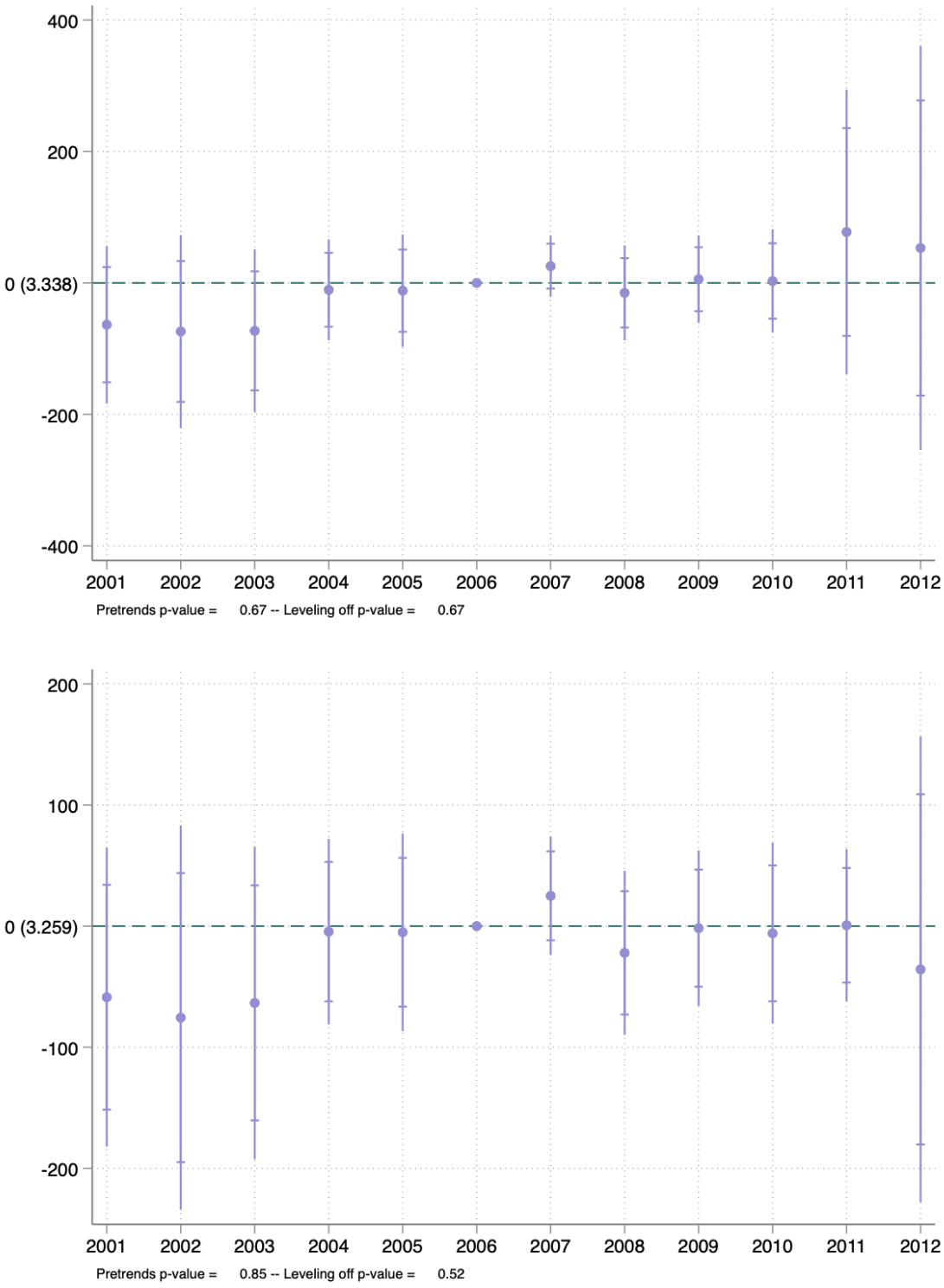


Figure 16: Product Type Level STC Event Study: Survival-Adjusted Entrant Firms. This figure plots the event study coefficients relative to 2006 from a product type level specification, where the outcome is the number of survival-adjusted entrants (refer to Section 4.5). The top panel includes all products; the bottom panel excludes phones. Exposure is constructed using a product type level probit. All regressions are weighted by the inverse variance of the number of products. Standard errors are clustered at the product type level.

Tables

Table 1: FCC Exhibit Confidentiality Rules. This table describes the long-term and short-term confidentiality eligibility for every type of exhibit in the FCC certification process. This table is directly from the Federal Communications Commission (2011).

Exhibit Type	Long-Term Confidentiality	Exceptions	Comments	Short-Term Confidentiality
Attestation Statements	No			No
Block Diagram	With Justification	Scanning Receivers (0.457(d)(i) – Always (no justification)); PCs, PC Peripherals, & unintentional radiators – No	Block diagrams for PCs, PC peripherals, other unintentional radiators not required by Rules therefore not held confidential; Short-Term Confidential N/A to scanning receivers	With Justification
Cover Letter(s)	No			No
External Photos	No			With Justification
ID Label / Location Info	No			No
Internal Photos	Special Circumstances (see comments)	Scanning Receivers (0.457(d)(i) – Always (no justification))	Epoxied / sealed circuit board, sealed device, non-consumer device serviceable only by Grantee or designated tech (non-disclosure agreement may be required)	With Justification
Operational Description	With Justification	Scanning Receivers (0.457(d)(i) – Always (no justification))	Short-Term Confidential N/A to scanning receivers	With Justification
Parts List / Tune Up Info	With Justification	Scanning Receivers (0.457(d)(i) – Always (no justification))	Short-Term Confidential N/A to scanning receivers	With Justification
RF Exposure Info	No			No
SDR Software / Security Info	Always		Long-Term Confidential always applies, therefore Short-Term Confidential N/A	N/A
Schematics	With Justification	Scanning Receivers (0.457(d)(i) – Always (no justification))	Short-Term Confidential N/A to scanning receivers	With Justification
Test Report	No			No
Test Setup Photos	No			With Justification
Users Manual	Special Circumstances (see comments)		"Technical" manual used as a service manual only by qualified tech (non-disclosure agreement may be required); or for use only by Other Equipment Manufacturer (OEM) and includes proprietary information	With Justification

Table 2: Variable Definitions from Product Attributes Dataset. This table lists and explains all explanatory variables in the analysis. The "Category" column indicates which framework the variable captures.

Product Attribute	Definition	Interpretation	Category
Number of rule parts	Low, Medium, or High number of regulatory rules.	High regulatory complexity generates additional costs and delays.	Regulatory Setting
Location (Country) — US/Canada firms	Home bias treatment from regulators		Regulatory Setting
EU firm	The presence of TCBs based in Europe gives these firms a favorable regulatory treatment.		Regulatory Setting
Asian firm	These firms face higher regulatory burdens.		Regulatory Setting
South American and Australian firms (omitted category)			Regulatory Setting
Firm filed products pre-2000	Established firm	Experience measure. Long-time incumbent who knows how to work the regulatory system and may have a valuable market brand.	Firms' Experience
Early Many Products	Firm has >20 products pre-2007	Experience measure. Long-time big producer with a brand that knows how to distribute many products.	Firms' Experience
New market	Product type emerged after 2001	Young market with high uncertainty and much exploratory activity.	Firms' Experience
Silicon Valley	Located in Silicon Valley city	Young, innovative firm with less experience and most likely venture-capital backing.	Firms' Experience
First-Time Entrant	The firm has never filed a product application before	Inexperienced with both meeting market and regulatory requirements, a lack of value, and a brand name.	Firms' Experience
Firm has one product ever	The firm filed only one product application.	Ex-post measure of the inexperienced firm with the low-quality product, a lack of value, and a brand name.	Firms' Experience
Firms with many total products	Firm has more than 200 lifetime products	Firm is large and experienced with FCC filings.	Firms' Experience
Firms with standard essential patents	Firms have filed with IEEE standards or are part of the Bluetooth SIG	Firms are experienced with the patent system and technology standard-setting.	Firms' Experience
Frequency ranges 1-2-3-4-5	Product complexity (lowest/low/medium /high/highest)	Technically complex product with high value added, increasing in complexity.	Value Capture
Has Bluetooth	Tests for operating in the 2.402 to 2.48 GHz frequency range	The product utilizes Bluetooth technology, which increases its complexity.	Value Capture
Bluetooth and SEP protection	Product operates Bluetooth and is part of Bluetooth SIG.	Product is protected by standard essential patent.	Value Capture
Has Wi-Fi	Tests for operating 2.412 to 2.472 GHz or 5.15 to 5.85 GHz frequency range	Product uses Wi-Fi, increasing complexity.	Value Capture
Wi-Fi and SEP protection	Product operates on Wi-Fi and is filed for IEEE 802.11	Product is protected by standard essential patent.	Value Capture
Wi-Fi and Bluetooth	Product tests for operating both Bluetooth and Wi-Fi frequency ranges	Product uses Wi-Fi and Bluetooth, increasing complexity.	Value Capture
Firm has many Wi-Fi SEPs	Firm has at least five filings under 802.11	Firm has high IPR protection in Wi-Fi.	Value Capture
Firm discloses Wi-Fi SEPs	Firm discloses patents in 802.11 filings	Firm has chosen to disclose technology patents.	Value Capture
Wide Market Presence	Firm spans >2 product classes.	Broad commercial aspirations: distributing products to many markets, and likely has a valuable brand.	Value Capture

Firm has >1 product in the year		Big commercial aspirations: distributing many products and likely has an established brand.	Value Capture
Firm has >5 products in the year		Firm producing many products and entering commodity status.	Value Capture
Outsourcing to Asia	US/Canada/EU firm uses Asian testing firm	Proxies for cost savings in response to competitive pressures, and proxies for a higher likelihood that the product has commodity status.	Competitive Pressure
Phone	Flag for mobile phone	The phone market is a complex product with numerous competitors and imitators.	Competitive Pressure
Modules and Transceivers	Flag for module, transceiver, boards, and chips	Simple commodity products are intermediate inputs.	Competitive Pressure
Market complex high	Complex product class	Many complex products offer many alternatives, increasing the potential imitators. Device operating in a complex market, which may increase the expected value.	Competitive Pressure
Market Consumer-oriented	Product type is the end market.	B2C and not B2B. Competitive Pressures shorten the product life cycles, raising competitive pressures.	Competitive Pressure
Competitive product class	Many (> 300) competing products within a product class	Many competitors and imitators increase competitive pressures for confidentiality.	Competitive Pressure
Heterogeneous product class	Product class complexity has high variance	Measures many indirect competitors at a broad level.	Competitive Pressure

Table 3: Average Variable Values Across Time Periods. This table presents the average values of explanatory variables in the data sample, separated by the three time periods of analysis: the early period (2001 to 2007), the transition period (2008 to 2010), and the late period (2011 to 2021).

Variables	2001 to 2007	2008 to 2010	2011 to 2021
No. of Ruleparts (capped)	1.183	1.418	1.374
US/Canada Firm	.388	.387	.3
EU Firm	.085	.105	.086
Asian Firm	.521	.499	.598
Firm filed product pre-2000	.527	.424	.197
Firm has >20 products pre-2007	.64	.597	.287
Product type emerge post-2001	.007	.015	.047
Silicon Valley	.035	.032	.027
Entrant Firm (< 1 year)	.257	.224	.28
Firm has one product ever	.048	.042	.076
Firm has more than 200 products	.27	.323	.174
Firm has any SEP (IEEE or SIG)	.084	.149	.071
Frequency Ranges (capped)	1.231	1.615	1.828
Bluetooth	.145	.254	.488
Bluetooth and SEP	.023	.072	.031
WiFi	.194	.223	.331
WiFi and SEP	.016	.034	.031
WiFi and Bluetooth	.015	.059	.162
Has more than 5 WiFi LOAs	.005	.014	.019
Discloses patents in WiFi LOAs	.014	.04	.038
Firm spans >2 product classes	.374	.428	.334
Firm has >1 product in year	.836	.846	.801
Firm has >5 products in year	.486	.507	.452
US/Canada firm test in Asia	.152	.185	.209
EU firm test in Asia	.017	.029	.039
Modules and Transceivers	.196	.185	.123
Phone	.078	.135	.088
Complex product class	.453	.649	.583
Competitive product class (>300)	.451	.55	.792
Product type is end market	.737	.735	.82
Heterogeneous product class	.527	.536	.621

Table 4: Count of Parts By Time Period. This table presents the percent of products which fall under each category of rule parts by the time periods of analysis: the early period (2001 to 2007), the transition period (2008 to 2010), and the late period (2011 to 2021). Note that products may fall under multiple subparts, and is counted in a category if any of the subparts abide. The total number of all products, regardless of rule part, is shown in the bottom row.

	2001 to 2007	2008 to 2010	2011 to 2021
Part 15	78%	81%	90%
Parts 15, 18, 22, 24, 27, 81, 87, 90, and 95	96%	95%	97%
Parts 21, 73, 74, 80, and 9	9%	6.7%	4.2%
Sample Size	33,784	20,396	127,555

Table 5: Tabulate of Confidentiality Usage at Product Level in 2011. We present the cross-tabulation of sample products using long-term and short-term confidentiality in a 2011, a representative year in the late period once usage has stabilized for both policies.

LTConfidentiality	STConfidentiality		
	0	1	Total
0	1,187	1	1,188
1	4,992	1,937	6,929
Total	6,179	1,938	8,117

Table 6: Tabulate of Confidentiality Usage for Firms with One Product in 2011. We present the cross-tabulation of sample one-time firms using long-term and short-term confidentiality in a 2011, a representative year in the late period once usage has stabilized for both policies. A one-time firm appears only once in our dataset, thus the number of products is equivalent to the number of firms.

LTConfidentiality	STConfidentiality		
	0	1	Total
0	294	0	294
1	952	145	1,097
Total	1,246	145	1,391

Table 7: Tabulate of Confidentiality Usage for Firms with Multiple Products in 2011. We present the cross-tabulation of sample firms using long-term and short-term confidentiality in a 2011, a representative year in the late period once usage has stabilized for both policies. A usage of 0.5 corresponds to partial usage of confidentiality by a firm across all of their products in 2011.

LTConfidentiality	STConfidentiality			
	0	0.5	1	Total
0	158	0	0	158
0.5	136	43	0	179
1	668	134	108	910
Total	962	177	108	1,247

Table 8: Probit of Product-Level LT Confidentiality from 2001 to 2007. This two-page table presents the probit regression coefficients of long-term confidentiality usage on explanatory variables using data from 2001 to 2007. Columns (1)-(4) only regress on subsets of the explanatory variables corresponding to regulatory setting, firm experience, value caption, and competition. Column (5) is our main specification which includes all explanatory variables. Column (6) is a robustness check which includes all Telecommunication Certification Body (TCB) fixed effects. We present the pseudo R-squared and cluster standard errors at the firm level.

VARIABLES	(1) Regulatory Setting	(2) Experience	(3) Value Capture	(4) Competitive	(5) All	(6) TCB Dummies
No. of Ruleparts (capped) = 2	0.895***				0.230**	0.170*
	(0.0881)				(0.102)	(0.102)
No. of Ruleparts (capped) = 3	0.503***				0.281*	0.307
	(0.117)				(0.171)	(0.187)
US/Canada Firm	-0.134				0.1000	0.0801
	(0.171)				(0.188)	(0.206)
EU Firm	0.370**				0.424**	0.138
	(0.175)				(0.199)	(0.224)
Asian Firm	-0.725***				-0.773***	-0.765***
	(0.162)				(0.179)	(0.200)
Firm filed product pre-2000		0.189**			0.0916	0.0752
		(0.0878)			(0.0736)	(0.0751)
Firm has >20 products pre-2007		0.0823			0.0344	-0.0226
		(0.0683)			(0.0599)	(0.0612)
Product type emerge post-2001		0.168			0.134	0.0759
		(0.116)			(0.116)	(0.123)
Silicon Valley		0.887***			0.446***	0.421***
		(0.120)			(0.136)	(0.126)
Entrant Firm (< 1 year)		0.0237			-0.0528	-0.0394
		(0.0507)			(0.0525)	(0.0555)
Firm has one product ever		-0.311***			-0.282***	-0.271***
		(0.0455)			(0.0539)	(0.0564)
Firm has more than 200 products		0.248**			0.222*	0.187
		(0.119)			(0.116)	(0.118)
Firm has any SEP (IEEE or SIG)		0.194			0.278	0.503***
		(0.227)			(0.248)	(0.154)
Observations	30,372	30,372	30,133	30,372	30,133	30,083
Pseudo R-sq	0.129	0.0844	0.199	0.125	0.319	0.413

VARIABLES	(1) Regulatory Setting	(2) Experience	(3) Value Capture	(4) Competitive	(5) All	(6) TCB Dummies
Frequency Ranges (capped) = 1			1.178***		1.132***	1.112***
			(0.159)		(0.127)	(0.136)
Frequency Ranges (capped) = 2			1.693***		1.291***	1.301***
			(0.146)		(0.132)	(0.128)
Frequency Ranges (capped) = 3			2.004***		1.426***	1.409***
			(0.181)		(0.172)	(0.179)
Frequency Ranges (capped) = 4			1.814***		1.333***	1.339***
			(0.188)		(0.186)	(0.207)
Frequency Ranges (capped) = 5			2.021***		1.528***	1.638***
			(0.243)		(0.265)	(0.303)
Bluetooth			0.724***		0.872***	0.737***
			(0.0654)		(0.0686)	(0.0692)
Bluetooth and SEP			0.772***		0.233	0.204
			(0.265)		(0.356)	(0.373)
WiFi			0.687***		0.871***	0.650***
			(0.0932)		(0.0897)	(0.0714)
WiFi and SEP			-1.002***		-1.266***	-1.294***
			(0.294)		(0.213)	(0.191)
WiFi and Bluetooth			-0.108		0.0776	0.218
			(0.207)		(0.245)	(0.282)
Has more than 5 WiFi LOAs			1.164***		0.602	0.579
			(0.409)		(0.376)	(0.439)
Discloses patents in WiFi LOAs			0.699**		0.360	0.124
			(0.323)		(0.274)	(0.254)
Firm spans >2 product classes			0.465***		0.186**	0.154**
			(0.0849)		(0.0732)	(0.0733)
Firm has >1 product in year			0.0721**		0.0165	0.0391
			(0.0321)		(0.0365)	(0.0383)
Firm has >5 products in year			-0.208***		-0.194***	-0.153**
			(0.0640)		(0.0592)	(0.0648)
US/Canada firm test in Asia				-0.129	-0.844***	-0.795***
				(0.117)	(0.115)	(0.106)
EU firm test in Asia				0.183	-1.053***	-1.007***
				(0.143)	(0.172)	(0.207)
Modules and Transceivers				-0.256***	-0.457***	-0.395***
				(0.0977)	(0.0897)	(0.0927)
Phone				1.496***	1.316***	1.286***
				(0.154)	(0.116)	(0.179)
Complex product class				-0.0441	-0.284***	-0.227***
				(0.0519)	(0.0506)	(0.0496)
Competitive product class (>300)				0.402***	0.596***	0.531***
				(0.0677)	(0.0674)	(0.0624)
Product type is end market				-0.629***	-0.479***	-0.368***
				(0.0840)	(0.0801)	(0.0833)
Heterogeneous product class				0.0319	-0.0572	-0.0336
				(0.0598)	(0.0551)	(0.0579)
Observations	30,372	30,372	30,133	30,372	30,133	30,083
Pseudo R-sq	0.129	0.0844	0.199	0.125	0.319	0.413

Table 9: Probit Marginal Effects of Product-Level LT Confidentiality from 2001 to 2007. This two-page table presents the marginal effect of each explanatory variable on the probability of long-term confidentiality usage using data from 2001 to 2007. Column (1) computes the marginal effect at the variable values of all observed data points then takes the average. Column (2) computes the marginal effect while holding all other variable values constant at the sample mean. Columns (3) and (4) are robustness checks that include all Telecommunication Certification Body (TCB) fixed effects. Standard errors are calculated using the delta method.

VARIABLES	(1) Observed	(2) At Mean	(3) Observed (TCB FE)	(4) At Mean (TCB FE)
No. of Ruleparts (capped) = 2	0.0558**	0.0701**	0.0361*	0.0503*
	(0.0240)	(0.0291)	(0.0215)	(0.0287)
No. of Ruleparts (capped) = 3	0.0677*	0.0842*	0.0642*	0.0860*
	(0.0394)	(0.0462)	(0.0378)	(0.0461)
US/Canada Firm	0.0248	0.0324	0.0172	0.0248
	(0.0466)	(0.0607)	(0.0443)	(0.0638)
EU Firm	0.105**	0.137**	0.0296	0.0426
	(0.0493)	(0.0643)	(0.0480)	(0.0692)
Asian Firm	-0.192***	-0.250***	-0.164***	-0.237***
	(0.0444)	(0.0582)	(0.0428)	(0.0618)
Firm filed product pre-2000	0.0227	0.0296	0.0162	0.0233
	(0.0183)	(0.0238)	(0.0161)	(0.0233)
Firm has >20 products pre-2007	0.00855	0.0111	-0.00485	-0.00699
	(0.0149)	(0.0194)	(0.0132)	(0.0190)
Product type emerge post-2001	0.0332	0.0433	0.0163	0.0235
	(0.0289)	(0.0376)	(0.0263)	(0.0379)
Silicon Valley	0.111***	0.144***	0.0905***	0.130***
	(0.0337)	(0.0441)	(0.0270)	(0.0392)
Entrant Firm (< 1 year)	-0.0131	-0.0171	-0.00847	-0.0122
	(0.0130)	(0.0170)	(0.0119)	(0.0171)
Firm has one product ever	-0.0701***	-0.0913***	-0.0581***	-0.0837***
	(0.0134)	(0.0174)	(0.0121)	(0.0175)
Firm has more than 200 products	0.0551*	0.0718*	0.0402	0.0578
	(0.0286)	(0.0371)	(0.0252)	(0.0363)
Firm has any SEP (IEEE or SIG)	0.0691	0.0900	0.108***	0.156***
	(0.0616)	(0.0804)	(0.0332)	(0.0476)
Frequency Ranges (capped) = 1	0.316***	0.425***	0.270***	0.413***
	(0.0334)	(0.0456)	(0.0342)	(0.0506)
Frequency Ranges (capped) = 2	0.357***	0.471***	0.312***	0.465***
	(0.0346)	(0.0448)	(0.0323)	(0.0467)
Frequency Ranges (capped) = 3	0.389***	0.506***	0.335***	0.491***
	(0.0435)	(0.0520)	(0.0427)	(0.0561)
Frequency Ranges (capped) = 4	0.367***	0.483***	0.320***	0.475***
	(0.0474)	(0.0571)	(0.0486)	(0.0628)
Frequency Ranges (capped) = 5	0.412***	0.530***	0.381***	0.537***
	(0.0617)	(0.0674)	(0.0629)	(0.0689)
Observations	30,133	30,133	30,083	30,083

VARIABLES	(1) Observed	(2) At Mean	(3) Observed (TCB FE)	(4) At Mean (TCB FE)
Bluetooth	0.217***	0.282***	0.158***	0.228***
	(0.0164)	(0.0232)	(0.0146)	(0.0222)
Bluetooth and SEP	0.0579	0.0755	0.0438	0.0631
	(0.0884)	(0.115)	(0.0801)	(0.115)
WiFi	0.216***	0.282***	0.140***	0.201***
	(0.0208)	(0.0304)	(0.0153)	(0.0237)
WiFi and SEP	-0.315***	-0.410***	-0.278***	-0.400***
	(0.0513)	(0.0692)	(0.0410)	(0.0604)
WiFi and Bluetooth	0.0193	0.0251	0.0468	0.0675
	(0.0608)	(0.0792)	(0.0604)	(0.0869)
Has more than 5 WiFi LOAs	0.150	0.195	0.124	0.179
	(0.0935)	(0.122)	(0.0943)	(0.136)
Discloses patents in WiFi LOAs	0.0893	0.116	0.0266	0.0383
	(0.0683)	(0.0891)	(0.0545)	(0.0786)
Firm spans >2 product classes	0.0461**	0.0601**	0.0330**	0.0476**
	(0.0181)	(0.0235)	(0.0156)	(0.0225)
Firm has >1 product in year	0.00409	0.00533	0.00840	0.0121
	(0.00905)	(0.0118)	(0.00821)	(0.0118)
Firm has >5 products in year	-0.0482***	-0.0629***	-0.0329**	-0.0474**
	(0.0147)	(0.0194)	(0.0139)	(0.0203)
US/Canada firm test in Asia	-0.210***	-0.273***	-0.171***	-0.246***
	(0.0263)	(0.0365)	(0.0213)	(0.0324)
EU firm test in Asia	-0.262***	-0.341***	-0.216***	-0.312***
	(0.0422)	(0.0553)	(0.0437)	(0.0633)
Modules and Transceivers	-0.114***	-0.148***	-0.0849***	-0.122***
	(0.0223)	(0.0292)	(0.0199)	(0.0289)
Phone	0.327***	0.426***	0.276***	0.398***
	(0.0288)	(0.0387)	(0.0373)	(0.0532)
Complex product class	-0.0704***	-0.0918***	-0.0487***	-0.0701***
	(0.0126)	(0.0164)	(0.0106)	(0.0154)
Competitive product class (>300)	0.148***	0.193***	0.114***	0.164***
	(0.0169)	(0.0220)	(0.0130)	(0.0191)
Product type is end market	-0.119***	-0.155***	-0.0791***	-0.114***
	(0.0200)	(0.0263)	(0.0179)	(0.0260)
Heterogeneous product class	-0.0142	-0.0185	-0.00721	-0.0104
	(0.0137)	(0.0178)	(0.0124)	(0.0179)
Observations	30,133	30,133	30,083	30,083

Table 10: Bivariate Probit of LT and ST Confidentiality from 2008 to 2010. This table presents the bivariate probit regression coefficients of long-term (left) and short-term (right) confidentiality usage on explanatory variables using data from 2008 to 2010, respectively. Columns (1)-(11) only regress on subsets of the explanatory variables corresponding to regulatory setting, firm experience, value caption, and competition. Columns (13) and (14) represent our main specification, which includes all explanatory variables. Columns (16) and (17) are a robustness check that includes all Telecommunication Certification Body (TCB) fixed effects. We cluster standard errors at the firm level and present the correlation between the error terms.

VARIABLES	(1) Regulatory Setting	(2) Regulatory Setting	(4) Experience	(5) Experience	(7) Value Capture	(8) Value Capture	(10) Competitive	(11) Competitive	(13) All Vars	(14) All Vars	(16) TCB Dummies	(17) TCB Dummies
Frequency Ranges (capped) = 1					0.794***	-0.358			0.758***	-0.350**	0.606***	-0.259
					(0.127)	(0.284)			(0.113)	(0.170)	(0.117)	(0.194)
Frequency Ranges (capped) = 2					1.009***	0.00129			0.961***	-0.270	0.767***	-0.151
					(0.137)	(0.291)			(0.122)	(0.189)	(0.128)	(0.214)
Frequency Ranges (capped) = 3					1.302***	0.273			1.105***	-0.301	0.958***	-0.140
					(0.152)	(0.260)			(0.158)	(0.207)	(0.171)	(0.228)
Frequency Ranges (capped) = 4					1.120***	0.470*			0.996***	-0.128	0.812***	0.0989
					(0.160)	(0.268)			(0.155)	(0.213)	(0.163)	(0.234)
Frequency Ranges (capped) = 5					0.913***	0.295			0.837***	-0.355	0.620***	-0.0809
					(0.169)	(0.262)			(0.166)	(0.225)	(0.178)	(0.248)
Bluetooth					0.0594	0.401***			0.165***	0.431***	0.119*	0.325***
					(0.0637)	(0.112)			(0.0636)	(0.0979)	(0.0620)	(0.0870)
Bluetooth and SEP					0.940***	0.392			0.717***	-0.221	0.476**	-0.251
					(0.175)	(0.412)			(0.211)	(0.259)	(0.224)	(0.227)
WiFi					0.141**	-0.202			0.137**	-0.0383	-0.0641	-0.0468
					(0.0651)	(0.135)			(0.0664)	(0.119)	(0.0715)	(0.118)
WiFi and SEP					-0.596*	0.445*			-0.452*	0.195	-0.563*	0.105
					(0.313)	(0.253)			(0.259)	(0.210)	(0.292)	(0.242)
WiFi and Bluetooth					0.400***	0.220			0.312**	0.195	0.552***	0.288*
					(0.140)	(0.149)			(0.156)	(0.157)	(0.177)	(0.159)
Has more than 5 WiFi LOAs					0.207	0.242			0.103	-0.231	0.259	-0.133
					(0.505)	(0.270)			(0.407)	(0.158)	(0.483)	(0.253)
Discloses patents in WiFi LOAs					0.566**	0.516**			0.532*	0.154	0.473	0.468*
					(0.239)	(0.258)			(0.282)	(0.268)	(0.313)	(0.272)
Firm spans >2 product classes					0.403***	0.354***			0.191**	0.0816	0.231**	0.0732
					(0.0836)	(0.137)			(0.0835)	(0.103)	(0.0899)	(0.0964)
Firm has >1 product in year					0.0719*	0.0316			0.0191	-0.00424	-0.00293	0.00931
					(0.0415)	(0.0560)			(0.0470)	(0.0649)	(0.0488)	(0.0632)
Firm has >5 products in year					0.183**	0.338***			0.151*	0.132	0.115	0.107
					(0.0816)	(0.0864)			(0.0853)	(0.0848)	(0.0865)	(0.0838)
Firm filed product pre-2000			0.206**	0.171					0.163*	0.138	0.130	0.0211
			(0.0915)	(0.171)					(0.0849)	(0.153)	(0.0847)	(0.132)
Firm has >20 products pre-2007			0.124	0.167					0.0482	0.0636	0.0437	0.111
			(0.0863)	(0.116)					(0.0899)	(0.113)	(0.0824)	(0.106)
Product type emerge post-2001			-0.194	0.176					-0.239*	0.193	-0.237*	0.261*
			(0.128)	(0.188)					(0.125)	(0.157)	(0.126)	(0.145)
Silicon Valley			0.431***	0.0475					0.439***	-0.190	0.420***	-0.174
			(0.151)	(0.184)					(0.135)	(0.167)	(0.138)	(0.163)
Entrant Firm (< 1 year)			-0.139**	-0.0659					-0.0907	-0.0313	-0.0196	-0.0515
			(0.0566)	(0.0735)					(0.0587)	(0.0780)	(0.0611)	(0.0774)
Firm has one product ever			-0.232***	-0.120					-0.215***	-0.172*	-0.243***	-0.191*
			(0.0598)	(0.0879)					(0.0668)	(0.101)	(0.0697)	(0.104)
Firm has more than 200 products			0.330***	0.633***					0.239*	0.360**	0.202	0.300**
			(0.117)	(0.199)					(0.129)	(0.162)	(0.140)	(0.146)
Firm has any SEP (IEEE or SIG)			0.330**	0.800***					-0.183	0.466**	-0.226	0.277
			(0.148)	(0.249)					(0.157)	(0.184)	(0.207)	(0.244)
No. of Ruleparts (capped) = 2	0.126*	0.437***							-0.334***	0.299***	-0.373***	0.151*
	(0.0713)	(0.0971)							(0.0777)	(0.0983)	(0.0838)	(0.0912)
No. of Ruleparts (capped) = 3	0.739***	1.425***							-0.166	0.788***	-0.270*	0.548***
	(0.106)	(0.182)							(0.132)	(0.153)	(0.149)	(0.154)
US/Canada Firm	0.317	1.074***							0.456**	0.631***	0.363*	0.460**
	(0.193)	(0.269)							(0.191)	(0.236)	(0.215)	(0.224)
EU Firm	0.443**	0.510**							0.515**	-0.00252	0.195	0.258
	(0.201)	(0.247)							(0.203)	(0.287)	(0.227)	(0.241)
Asian Firm	-0.104	0.571**							-0.331*	0.149	-0.467**	0.0290
	(0.193)	(0.238)							(0.178)	(0.221)	(0.198)	(0.218)
US/Canada firm test in Asia							-0.0580	0.392*	-0.730***	-0.120	-0.767***	-0.152
							(0.0672)	(0.229)	(0.100)	(0.138)	(0.115)	(0.127)
EU firm test in Asia							-0.178	-0.0619	-0.831***	0.0320	-0.606***	0.0210
							(0.123)	(0.194)	(0.141)	(0.169)	(0.162)	(0.155)
Modules and Transceivers							-0.374***	-0.0708	-0.371***	-0.0937	-0.417***	-0.0188
							(0.121)	(0.178)	(0.118)	(0.157)	(0.136)	(0.142)
Phone							0.651***	1.138***	0.226*	-0.0172	0.386**	-0.0463
							(0.181)	(0.323)	(0.128)	(0.231)	(0.154)	(0.206)
Complex product class							-0.316***	-0.246**	-0.337***	-0.215***	-0.250***	-0.235***
							(0.0702)	(0.0981)	(0.0793)	(0.0619)	(0.0772)	(0.0641)
Competitive product class (>300)							0.0101	-0.140	0.0857	-0.0933	-0.0450	-0.143
							(0.109)	(0.1000)	(0.101)	(0.110)	(0.114)	(0.0981)
Product type is end market							-0.279***	-0.172	-0.176**	-0.260**	-0.137	-0.181
							(0.0837)	(0.132)	(0.0810)	(0.130)	(0.0880)	(0.118)
Heterogeneous product class							0.663***	0.524***	0.486***	0.210**	0.532***	0.219**
							(0.0980)	(0.108)	(0.0902)	(0.0907)	(0.0939)	(0.0919)
Observations	18,462	18,462	18,462	18,462	18,385	18,385	18,462	18,462	18,385	18,385	18,385	18,385
Error Correlation	0.794	0.794	0.781	0.781	0.815	0.815	0.795	0.795	0.815	0.815	0.860	0.860

Table 11: Probit of Product-Level ST Confidentiality from 2011 Onwards. This two-page table presents the probit regression coefficients of short-term confidentiality usage on explanatory variables using data from 2011 to 2021. Columns (1)-(4) only regress on subsets of the explanatory variables corresponding to regulatory setting, firm experience, value caption, and competition. Column (5) is our main specification which includes all explanatory variables. Column (6) is a robustness check which includes all Telecommunication Certification Body (TCB) fixed effects. We present the pseudo R-squared and cluster standard errors at the firm level.

VARIABLES	(1) Regulatory Setting	(2) Experience	(3) Value Capture	(4) Competitive	(5) All	(6) TCB Dummies
No. of Ruleparts (capped) = 2	0.506***				0.240***	0.210***
	(0.0362)				(0.0494)	(0.0467)
No. of Ruleparts (capped) = 3	0.860***				0.317***	0.337***
	(0.115)				(0.0912)	(0.0816)
US/Canada Firm	0.840***				0.523***	0.530***
	(0.180)				(0.146)	(0.135)
EU Firm	0.711***				0.360**	0.326**
	(0.145)				(0.156)	(0.145)
Asian Firm	0.376***				0.0206	-0.0263
	(0.144)				(0.141)	(0.129)
Firm filed product pre-2000		0.256*			0.152	0.0539
		(0.135)			(0.121)	(0.107)
Firm has >20 products pre-2007		0.275***			0.263***	0.227**
		(0.101)			(0.0961)	(0.0911)
Product type emerge post-2001		0.0780			0.166***	0.136**
		(0.0632)			(0.0627)	(0.0639)
Silicon Valley		0.596***			0.384***	0.389***
		(0.109)			(0.113)	(0.0973)
Entrant Firm (< 1 year)		-0.228***			-0.187***	-0.174***
		(0.0334)			(0.0304)	(0.0303)
Firm has one product ever		0.0259			0.0273	0.0313
		(0.0265)			(0.0304)	(0.0309)
Firm has more than 200 products		0.571***			0.553***	0.502***
		(0.138)			(0.126)	(0.122)
Firm has any SEP (IEEE or SIG)		0.682***			0.0652	0.126
		(0.176)			(0.193)	(0.188)
Observations	119,309	119,309	119,048	119,309	119,048	118,849
Pseudo R-sq	0.0786	0.144	0.135	0.0431	0.192	0.261

VARIABLES	(1) Regulatory Setting	(2) Experience	(3) Value Capture	(4) Competitive	(5) All	(6) TCB Dummies
Frequency Ranges (capped) = 1			-0.116		-0.174	-0.277
			(0.261)		(0.226)	(0.234)
Frequency Ranges (capped) = 2			0.145		-0.0614	-0.161
			(0.273)		(0.235)	(0.241)
Frequency Ranges (capped) = 3			0.220		-0.159	-0.274
			(0.268)		(0.252)	(0.251)
Frequency Ranges (capped) = 4			0.442*		-0.0128	-0.127
			(0.251)		(0.238)	(0.245)
Frequency Ranges (capped) = 5			0.590**		0.0401	-0.0759
			(0.243)		(0.248)	(0.247)
Bluetooth			-0.0595		0.0846**	0.101***
			(0.0415)		(0.0387)	(0.0370)
Bluetooth and SEP			1.004***		0.532**	0.321*
			(0.131)		(0.211)	(0.180)
WiFi			-0.241***		-0.130***	-0.120***
			(0.0470)		(0.0484)	(0.0457)
WiFi and SEP			0.615***		0.522***	0.417***
			(0.157)		(0.160)	(0.145)
WiFi and Bluetooth			0.266***		0.237***	0.204***
			(0.0639)		(0.0586)	(0.0567)
Has more than 5 WiFi LOAs			0.203		0.136	0.229*
			(0.128)		(0.162)	(0.134)
Discloses patents in WiFi LOAs			0.332		0.0261	-0.178
			(0.204)		(0.237)	(0.207)
Firm spans >2 product classes			0.441***		0.171***	0.155***
			(0.0829)		(0.0601)	(0.0593)
Firm has >1 product in year			0.0320		0.00256	0.0181
			(0.0221)		(0.0233)	(0.0228)
Firm has >5 products in year			0.0601		-0.0994**	-0.0662
			(0.0567)		(0.0421)	(0.0404)
US/Canada firm test in Asia				0.266**	-0.375***	-0.457***
				(0.110)	(0.0727)	(0.0714)
EU firm test in Asia				0.230***	0.0467	-0.0127
				(0.0878)	(0.0904)	(0.0968)
Modules and Transceivers				-0.125	-0.0633	-0.0624
				(0.0923)	(0.0774)	(0.0683)
Phone				0.648***	0.00657	0.0839
				(0.139)	(0.109)	(0.0994)
Complex product class				0.131***	0.0370	0.0139
				(0.0387)	(0.0352)	(0.0376)
Competitive product class (>300)				-0.0996***	-0.0925***	-0.0753**
				(0.0386)	(0.0322)	(0.0321)
Product type is end market				-0.287***	-0.211***	-0.154**
				(0.0804)	(0.0794)	(0.0698)
Heterogeneous product class				0.152***	0.0146	0.0227
				(0.0507)	(0.0451)	(0.0456)
Observations	119,309	119,309	119,048	119,309	119,048	118,849
Pseudo R-sq	0.0786	0.144	0.135	0.0431	0.192	0.261

Table 12: Probit Marginal Effects of Product-Level ST Confidentiality from 2011 Onwards. This two-page table presents the marginal effect of each explanatory variable on the probability of short-term confidentiality usage using data from 2011 to 2021. Column (1) computes the marginal effect at the variable values of all observed data points then takes the average. Column (2) computes the marginal effect while holding all other variable values constant at the sample mean. Columns (3) and (4) are robustness checks that include all Telecommunication Certification Body (TCB) fixed effects. Standard errors are calculated using the delta method.

VARIABLES	(1) Observed	(2) At Mean	(3) Observed (TCB FE)	(4) At Mean (TCB FE)
No. of Ruleparts (capped) = 2	0.0638***	0.0733***	0.0502***	0.0612***
	(0.0138)	(0.0157)	(0.0117)	(0.0141)
No. of Ruleparts (capped) = 3	0.0860***	0.0992***	0.0836***	0.103***
	(0.0267)	(0.0306)	(0.0219)	(0.0268)
US/Canada Firm	0.131***	0.154***	0.122***	0.150***
	(0.0366)	(0.0431)	(0.0309)	(0.0381)
EU Firm	0.0904**	0.106**	0.0749**	0.0924**
	(0.0391)	(0.0459)	(0.0332)	(0.0410)
Asian Firm	0.00517	0.00604	-0.00603	-0.00744
	(0.0355)	(0.0415)	(0.0296)	(0.0366)
Firm filed product pre-2000	0.0382	0.0447	0.0124	0.0153
	(0.0303)	(0.0355)	(0.0244)	(0.0301)
Firm has >20 products pre-2007	0.0660***	0.0772***	0.0521**	0.0643**
	(0.0241)	(0.0284)	(0.0208)	(0.0259)
Product type emerge post-2001	0.0418***	0.0488***	0.0312**	0.0386**
	(0.0158)	(0.0186)	(0.0147)	(0.0182)
Silicon Valley	0.0964***	0.113***	0.0892***	0.110***
	(0.0284)	(0.0330)	(0.0223)	(0.0274)
Entrant Firm (< 1 year)	-0.0470***	-0.0549***	-0.0400***	-0.0494***
	(0.00786)	(0.00912)	(0.00714)	(0.00882)
Firm has one product ever	0.00685	0.00800	0.00718	0.00886
	(0.00765)	(0.00893)	(0.00710)	(0.00875)
Firm has more than 200 products	0.139***	0.162***	0.115***	0.142***
	(0.0316)	(0.0384)	(0.0281)	(0.0356)
Firm has any SEP (IEEE or SIG)	0.0164	0.0191	0.0288	0.0356
	(0.0486)	(0.0567)	(0.0432)	(0.0532)
Frequency Ranges (capped) = 1	-0.0454	-0.0524	-0.0678	-0.0838
	(0.0621)	(0.0725)	(0.0616)	(0.0780)
Frequency Ranges (capped) = 2	-0.0166	-0.0193	-0.0408	-0.0508
	(0.0646)	(0.0754)	(0.0633)	(0.0799)
Frequency Ranges (capped) = 3	-0.0419	-0.0484	-0.0672	-0.0831
	(0.0686)	(0.0800)	(0.0652)	(0.0823)
Frequency Ranges (capped) = 4	-0.00352	-0.00410	-0.0325	-0.0406
	(0.0654)	(0.0762)	(0.0643)	(0.0810)
Frequency Ranges (capped) = 5	0.0112	0.0131	-0.0197	-0.0247
	(0.0687)	(0.0799)	(0.0651)	(0.0820)
Observations	119,048	119,048	118,849	118,849

VARIABLES	(1) Observed	(2) At Mean	(3) Observed (TCB FE)	(4) At Mean (TCB FE)
Bluetooth	0.0212**	0.0248**	0.0233***	0.0287***
	(0.00972)	(0.0114)	(0.00848)	(0.0105)
Bluetooth and SEP	0.134**	0.156**	0.0737*	0.0909*
	(0.0529)	(0.0620)	(0.0411)	(0.0506)
WiFi	-0.0327***	-0.0382***	-0.0274***	-0.0338***
	(0.0122)	(0.0143)	(0.0105)	(0.0130)
WiFi and SEP	0.131***	0.153***	0.0957***	0.118***
	(0.0402)	(0.0473)	(0.0332)	(0.0413)
WiFi and Bluetooth	0.0595***	0.0696***	0.0468***	0.0578***
	(0.0148)	(0.0174)	(0.0131)	(0.0162)
Has more than 5 WiFi LOAs	0.0341	0.0398	0.0525*	0.0648*
	(0.0408)	(0.0475)	(0.0308)	(0.0378)
Discloses patents in WiFi LOAs	0.00657	0.00768	-0.0407	-0.0503
	(0.0595)	(0.0696)	(0.0476)	(0.0589)
Firm spans >2 product classes	0.0430***	0.0503***	0.0355***	0.0438***
	(0.0151)	(0.0178)	(0.0137)	(0.0169)
Firm has >1 product in year	0.000642	0.000751	0.00415	0.00512
	(0.00586)	(0.00685)	(0.00524)	(0.00646)
Firm has >5 products in year	-0.0250**	-0.0292**	-0.0152*	-0.0187*
	(0.0104)	(0.0122)	(0.00919)	(0.0113)
US/Canada firm test in Asia	-0.0941***	-0.110***	-0.105***	-0.129***
	(0.0182)	(0.0209)	(0.0163)	(0.0199)
EU firm test in Asia	0.0117	0.0137	-0.00292	-0.00360
	(0.0227)	(0.0265)	(0.0222)	(0.0274)
Modules and Transceivers	-0.0159	-0.0186	-0.0143	-0.0176
	(0.0195)	(0.0228)	(0.0157)	(0.0194)
Phone	0.00165	0.00193	0.0192	0.0238
	(0.0273)	(0.0320)	(0.0228)	(0.0282)
Complex product class	0.00929	0.0109	0.00319	0.00394
	(0.00883)	(0.0104)	(0.00862)	(0.0107)
Competitive product class (>300)	-0.0232***	-0.0272***	-0.0173**	-0.0213**
	(0.00806)	(0.00953)	(0.00736)	(0.00913)
Product type is end market	-0.0529***	-0.0618***	-0.0354**	-0.0437**
	(0.0200)	(0.0235)	(0.0160)	(0.0198)
Heterogeneous product class	0.00366	0.00427	0.00521	0.00643
	(0.0113)	(0.0132)	(0.0105)	(0.0129)
Observations	119,048	119,048	118,849	118,849

A1 Regulatory Rule Parts

The rules in Part 15 cover devices that use unlicensed spectrum, as do Parts 18, 95, 96, and 97. However, Part 15 is far from the only device category shaped by the rule changes. There are many others. The FCC Part 18 rules regulate industrial, scientific, and medical (ISM) equipment. The FCC Part 95 rules regulate Personal Radio Services, which include license-free and low-power two-way radio communications for individuals and businesses. Part 96 – Covers the Citizens Broadband Radio Service (CBRS) in the 3.5 GHz band (3550-3700 MHz), which includes both licensed and unlicensed (General Authorized Access) users. Part 97 (in some cases) governs Amateur Radio (ham radio).

Many cellular and smartphone devices follow the rules defined in Parts 20, 22, 24, 27, 90, and 96 of the FCC regulations. Part 15 primarily covers consumer electronics, smart home devices, the Internet of Things (IoT), and networking products. Part 20 covers commercial mobile radio services (CMRS), which are mobile services interconnected with the Public Switched Telephone Network (PSTN). The FCC Part 22 rules regulate Public Mobile Services, which include commercial cellular networks, paging services, and other mobile radio communication systems. The FCC Part 24 rules regulate Personal Communications Services (PCS), a wireless communications category operating in the 1850–1990 MHz frequency range. The FCC Part 27 rules govern various services, most prominently the 5G and LTE services. Part 90 covers land mobile radio communications for non-commercial and critical infrastructure use cases. Part 96 is CBRS, as described above. Relatedly, Part 15 devices are sold in commercial markets, where devices that satisfy other parts are also sold. For example, Parts 18, 27, 95, 96, and 90 regulate devices often sold in similar commercial markets, including enterprise wireless, industrial automation, retail communications, and professional audio/visual equipment.

The FCC's rule changes also affected additional Parts, including Parts 21, 73, 74, 80, and 9, which this study does not focus on. The FCC Part 21 rules governed the licensing and operation of domestic public fixed and mobile microwave services, including point-to-point and point-to-multipoint microwave communications. The FCC Part 73 rules regulate broadcast radio and television services, applying to AM, FM, and television broadcasting (both analog and digital). The FCC Part 74 rules regulate auxiliary broadcast services, such as wireless microphones, electronic news gathering (ENG), and studio-to-transmitter links (STL). The FCC Part 80 rules regulate maritime communications, covering radio operations on ships, coastal stations, and other naval communication systems. The FCC Part 87 rules regulate Aviation Services, covering radio communication for aircraft, airports, air traffic control, and aeronautical navigation. The FCC Part 9 rules regulate interconnected Voice over Internet Protocol (VoIP) services and broadband Internet access services (BIAS), ensuring compliance with 911, public safety, accessibility, and network reliability requirements. Part 9 focuses on VoIP and broadband services, while Part 90 covers land mobile radio communications for non-commercial and critical infrastructure use cases.

Table A.1: Product Code Hierarchy. This appendix table presents the hierarchical product classification system developed for this study, covering device applications in the FCC equipment authorization database from 1982 to 2021. The three-level taxonomy comprises 26 Classes (shown in bold), 83 Families (shown in italics), and 298 Types.

Product Code Hierarchy

Class	*Family*	Type
Audio Equipment	*Audio Equipment*	Amplifiers; Compact disc players; Home stereo systems; Home tape recorders; Portable stereo systems
	Microphones	Microphones
	Speakers	Speaker systems; Jukeboxes
Camera	*Aerial Cameras*	Aerial cameras; Drone
	Baby Monitor Camera	Baby monitor camera
	Digital Camera	Digital cameras
	Motor Vehicle Camera	Motor vehicle camera
	Security Camera	Security camera
	Video Camera	Camcorders; Video camera; Video cameras (except household-type)
Camera Equipment	*Camera Equipment*	Light meters; Microfilm equipment; Night vision optical devices; Photographic equipment; Range finders, photographic
Computer	*Desktop Computer*	Minicomputers; Personal computers
	Portable Computer	Laptop computers; Notebook computers; Portable computers
	Tablet	Tablet computers
Computer Input/Output Equipment	*Baby Monitor*	Baby monitor
	Biometric Input Device	Biometrics system input devices
	Card Reader	Smart card readers
	Computer Input/Output Equipment	Bar code scanners; Computer input/output equipment; Joystick devices; Optical readers and scanners; Plotters, computer
	Flat Panel Displays	Flat panel displays
	Keyboard	Keyboards, computer peripheral equipment
	Monitor	Monitors, computer peripheral equipment
	Mouse	Mouse
	Printer	Printers, computer
Computer Storage Device	*Computer Storage Device*	CD-ROM drives; Disk drives, computer; Floppy disk drives; Hard disk drives, computer peripheral equipment; Optical disk drives; Storage devices, computer; Tape storage units (e.g., drive backups), computer peripheral equipment; USB flash drives (e.g., thumbdrives, jumpdrives)

Product Code Hierarchy (Continued)

Class	*Family*	Type
Computer Terminal/Peripheral	*Computer Terminal/Peripheral*	Computer terminal/peripheral
	Point of Sale Terminals	Point of sale terminals
Data Communications Equipment	*Data Communications Equipment*	Data communications equipment; Private branch exchange (PBX) equipment; Repeater and transceiver equipment
	Facsimile Equipment	Facsimile equipment
	Modem	Modem
Electrical Appliance	*Electrical Appliance*	Air purification equipment; Blenders; Ceiling fans; Coffee makers; Cooking appliances; Deep-fat fryers; Dehumidifiers; Desks (except wood), office-type; Dishwashers; Dry shavers; Electric blankets; Electric heating pads; Hair clippers for human use; Hot water heaters; Hotplates; Humidifiers; Massage machines; Microwave ovens; Refrigerators; Toilet; Toothbrushes, electric; Vacuum cleaners
Headset	*Earbuds*	Earbuds
	Headphones	Headphones
	Headset	Headset
Intercom Equipment	*Intercom Equipment*	Intercom systems and equipment
Lighting Equipment	*Commercial Lighting Equipment*	Commercial lighting fixtures
	Other Lighting Equipment	Christmas tree lighting sets; Compact fluorescent light bulbs; Electric light bulbs; Fireplace logs, electric; Flashlight bulb; Floodlights; Lanterns; Spotlights; Street lighting fixtures; Ultraviolet lamp fixtures; Underwater lighting fixtures
	Residential Lighting Equipment	Ceiling lighting fixtures; Floor lamps, residential; Pendant lamps fixtures; Residential lighting fixtures; Table lamps; Wall lamps; Yard lights
Locking Devices	*Door Locking Devices*	Door locking devices
	Locking Devices	Locking devices

Product Code Hierarchy (Continued)

Class	*Family*	Type
Measure/Display/ Control Instruments	*Environmental Control System*	Computerized environmental control systems; Energy cutoff controls, residential and commercial types
	Industrial Controls	Control equipment, electric
	Laboratory Instrument	Microscopes, electron and proton; Spectrometers; Spectrophotometers
	Measure/Display/Control Instruments	Accelerometers; Breathalyzers; Density and specific gravity instruments; Electric and electronic controllers; Fluidic devices, circuits, and systems for process control; Gas and liquid analysis instruments; Hygrometers; Industrial process control instruments; Infrared instruments; Instruments for industrial process control; Level and bulk measuring instruments; Level gauges; Light responsive appliance controls; Manometers, industrial process-type; Measuring instruments; Metal detectors; Moisture meters; Pressure gauges; Rain gauges; Testing equipment; Thermometer; Thermometers, filled system industrial process-type; Time clocks and time recording devices
	Meter	Gasmeters, large capacity, domestic and industrial; Gauges; Meters; Parking meters; Pedometers; Tallying meters; Watermeters
	Temperature Controls	Appliance controls; Heating regulators; Space heaters; Temperature instruments; Thermostats
Medical Equipment	*Medical Equipment*	Audiological equipment; Blood pressure apparatus; Cardiographs; Defibrillators; Electromedical equipment; Electromyographs; Electrosurgical units; Endoscopic equipment, electromedical; Hearing aids, electronic; Magnetic resonance imaging (MRI) medical diagnostic equipment; Otoscopes, electromedical equipment; Pacemakers; Stethoscopes; Thermometers, medical; Ultrasonic medical equipment; X-ray apparatus and tubes manufactuing
Microprocessor Chip	*Microprocessor Chip*	Microprocessor chip
Miscellaneous	*Miscellaneous*	Air-conditioning equipment; Amusement machines, coin- or card-operated; Animal traps; Cash registers (except point of sales terminals); Coolers, refrigeration; Handheld label making equipment; Miscellaneous; Soda fountain cooling and dispensing equipment; Telescopes; Typewriters; Voting machines
	Musical Equipment	Electric musical instruments
	Pet Equipment	Pet equipment
	Scale	Scales, including laboratory-type
	Traffic Systems	Traffic advisory and signalling systems

Product Code Hierarchy (Continued)

Class	*Family*	Type
Mobile Communications Equipment	*Antennas*	Antennas, transmitting and receiving
	Location Positioning Equipment	Beacon; Global positioning system (GPS) equipment; Location positioning equipment; Motor vehicle GPS/navigation; Tag tech
	Mobile Communications Equipment	Broadcast equipment; Earth station communications equipment; Marine radio communications equipment; Mobile communications equipment; Pagers; Satellite communications equipment
Module	*Board*	Board
	Card	Card
	Module	Module
Motor Vehicle Equipment	*Bikes and Scooters*	Bicycles and parts; Motor scooters; Motorcycles and parts
	Keyless Entry Systems	Keyless entry systems
	Motor Vehicle Media System	Automobile radio receivers; Car stereos; Motor vehicle media system
	Motor Vehicle Monitoring System	Automotive diagnostic equipment; Electrical control chips (modules), motor vehicle; Electrical ignition cable sets for internal combustion engines; Instrument control panels
	Motor Vehicle Security System	Automotive theft alarm systems; Motor vehicle security system
Phone	*Landline Phone*	Cordless telephones (except cellular); Telephone carrier line equipment
	Mobile Phone	Cellular telephones (new and rebuilt); Smartphones (new and rebuilt)
Power Flow Equipment	*Battery Chargers*	Battery chargers
	Current-Carrying Wiring Device	Outlets; Plugs; Sockets
	Motors	Motors, electric; Power generators
	Power Flow Equipment	Inverters, solid-state; Power converter units (i.e., AC to DC); Surge suppressors; Uninterruptible power supplies (UPS)
	Power Flow Measuring Instruments	Battery testers; Current measuring equipment; Energy measuring equipment; Multimeters; Oscillators; Oscilloscopes; Pulse (i.e., signal) generators; Spectrum analyzers; Test equipment for electronic and electrical circuits and equipment
	Switches	Circuit breakers; Dimmer switches; Switches for electrical wiring; Switches, electric power

Product Code Hierarchy (Continued)

Class	*Family*	Type
Radar Equipment	*Radar Equipment*	Fish finders; Radar systems and equipment
Radio Equipment	*Radio Equipment*	Airborne radio communications equipment; Citizens' band (CB) radios; Clock radios; Radio receiving sets; Radio transmitting antennas and ground equipment
Remote Control	*Remote Control*	Remote control
RFID Devices	*RFID Devices*	Radio frequency identification (RFID) devices
Search/Navigation Equipment	*Search/Navigation Equipment*	Aeronautical systems and instruments; Altimeters
Security Equipment	*Alarm Systems*	Alarm systems and equipment; Motion alarms (e.g., swimming pool, perimeter); Sirens (e.g., air raid, industrial, marine, vehicle)
	Security Equipment	Carbon monoxide detectors; Glass breakage detection and signaling devices; Motion detectors, security system
	Smoke Detectors	Smoke detectors
	Theft Prevention Devices	Theft prevention signaling devices
Sports Equipment	*Sports Equipment*	Fitness equipment; Golfing equipment; Skateboards
Toy	*Toy*	Children's gocarts; Children's scooters; Electronic toys and games
Transceiver	*Transceiver*	Transceiver
Video Equipment	*Media Player*	Digital video disc players; Home tape players; Media player
	Projector	Projection equipment; Video projectors, household-type
	Television Equipment	Cable decoder; Television (TV) sets
	Video Equipment	E-book readers; Home theater audio and video equipment; Video cassette recorders (VCR)
Video-Game Equipment	*Video-Game Equipment*	Gaming consoles; Video game devices
Wearable Equipment	*Personal Safety Devices*	Helmets, athletic; Personal safety devices; Respiratory protection mask
	Wearable Equipment	Fitness wearable equipment; Smartwatches; Wearable equipment

A2 Regulatory Capture of TCBs

Since TCBs file for confidentiality on behalf of firms, the ultimate form of regulatory capture in this setting might be the knowledge that specific TCBs possess. To account for the most extreme case, Tables 8 through 12 include a specification with TCB fixed effects to soak up any TCB specific knowledge. Reassuringly, the probit coefficients do not significantly change with the inclusion of TCB fixed effects. There is a total of 45 distinct TCBs in our twenty-year sample.

It is also possible that TCBs with closer ties to the FCC may have systematically higher rates of confidentiality usage, controlling for product and market-related variables. Figures A.1 and A.2 show the raw usage of long-term and short-term confidentiality for the top ten TCBs in each of the three time periods. Between 2001 and 2007, three TCBs (Timco, UL, & ACB) dominated the certification market, exhibiting heterogeneity in long-term confidentiality usage. Starting in 2008, there is also the presence of short-term confidentiality (we almost never see cases of only short-term opt-in without long-term). Levels of short-term confidentiality were overall low from 2008 to 2010, except for PCTEST Engineering, a TCB created by former FCC engineers that specializes in highly complex products. Post-2011, rates of long-term and short-term confidentiality are similar across the top TCBs.

We see a similar progression in the manufacturer location breakdown in Figure A.3. In 2001 to 2007, there is more heterogeneity in manufacturer location across TCBs, with PCTEST Engineering and the FCC focusing on domestic (US and Canada) firms. Post-2011, the breakdown of manufacturer location is much more homogeneous across the top TCBs. Overall, we do not find significant evidence of specialized knowledge transmission of confidentiality protocols to certain TCBs. This includes non-US TCBs based in Europe.

Appendix Figures

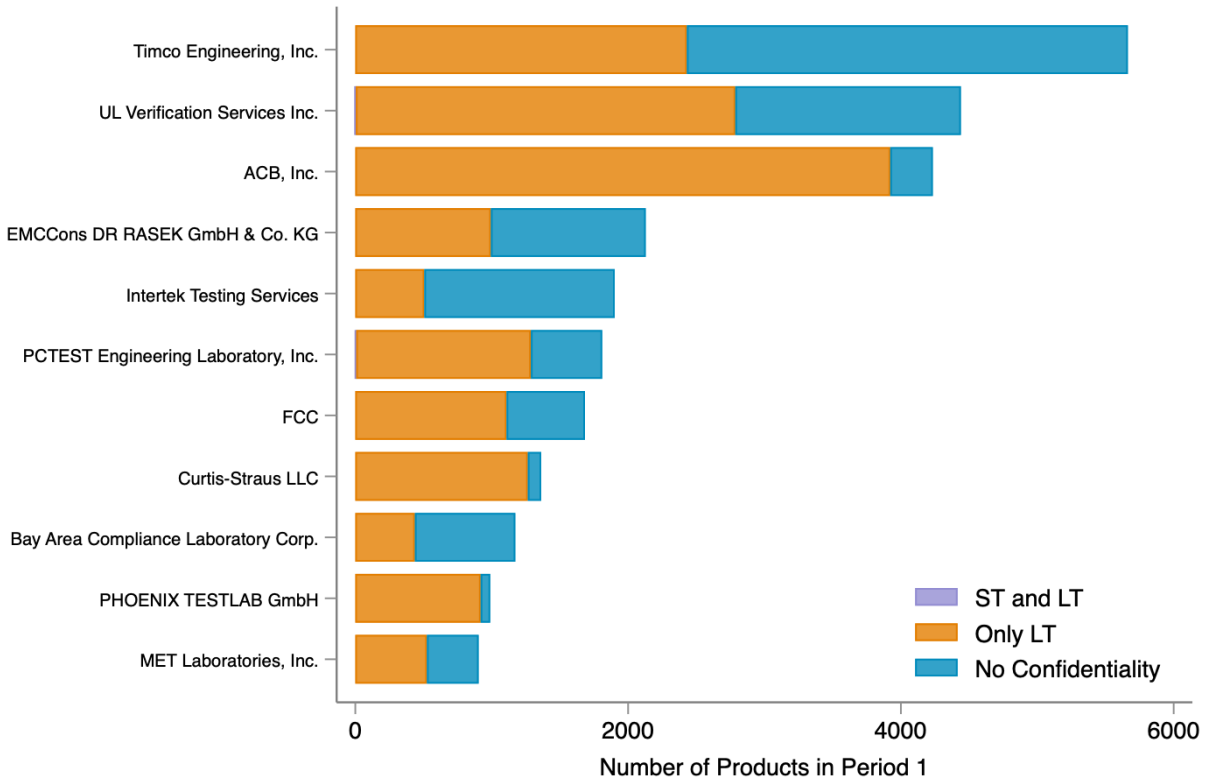


Figure A.1: Confidentiality Usage of Top 10 TCBs in 2001 to 2007. This figure plots total number of certified products from 2001 to 2007 of the top ten Telecommunication Certification Bodies (TCBs), broken down by confidentiality usage. The blue bar represents no confidentiality usage, the orange bar represents only long-term confidentiality, and the purple bar represents both long-term (LT) and short-term (ST) confidentiality usage. Note that short-term confidentiality did not exist in this early period.

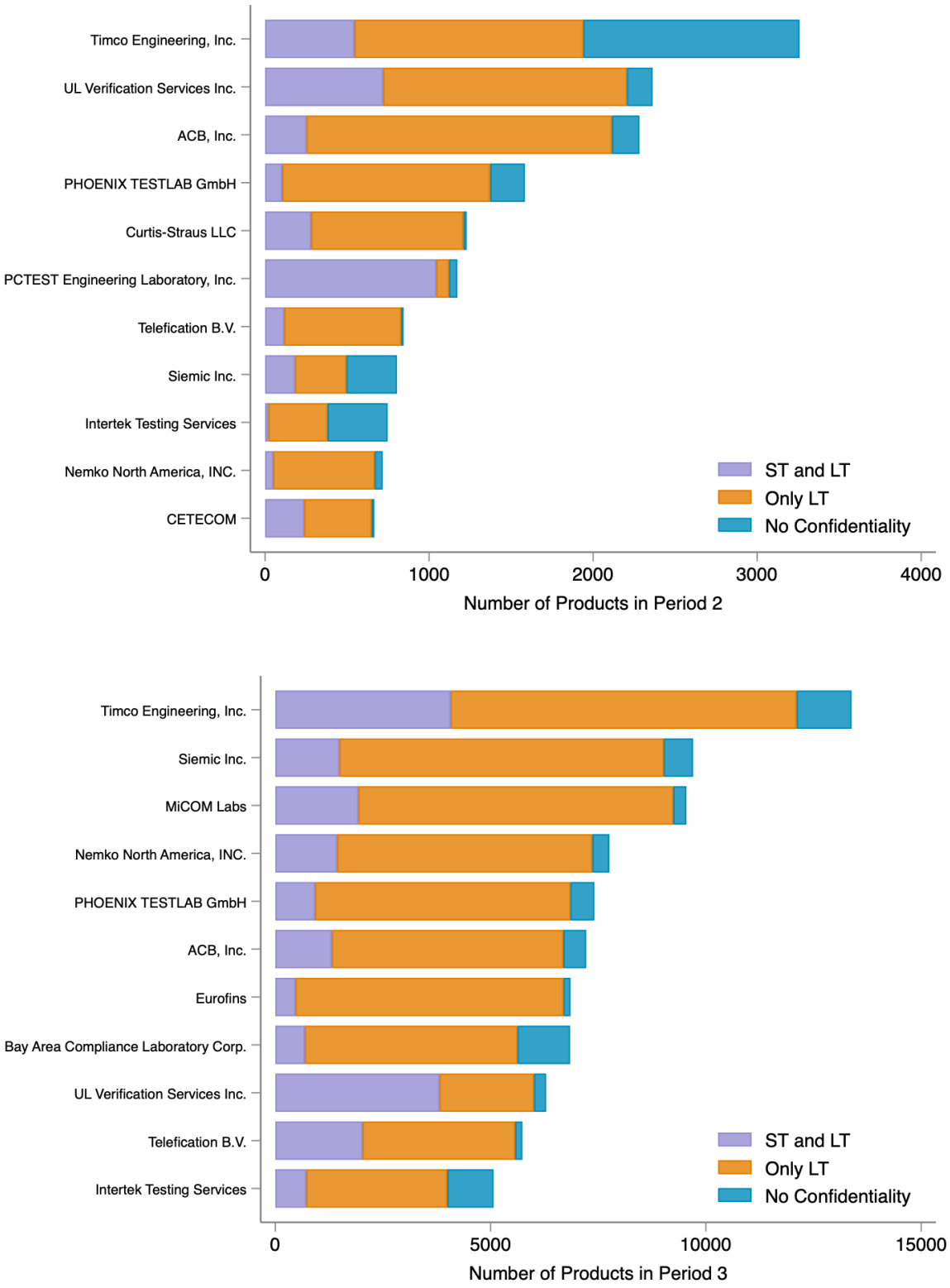


Figure A.2: Confidentiality Usage of Top 10 TCBs in 2008 to 2010 and 2011 to 2021. This figure plots total number of certified products of the top ten Telecommunication Certification Bodies (TCBs), broken down by confidentiality usage. The top and bottom panels plot all products in 2008 to 2010 and 2011 to 2021, respectively. The blue bar represents no confidentiality usage, the orange bar represents only long-term confidentiality, and the purple bar represents both long-term (LT) and short-term (ST) confidentiality usage.

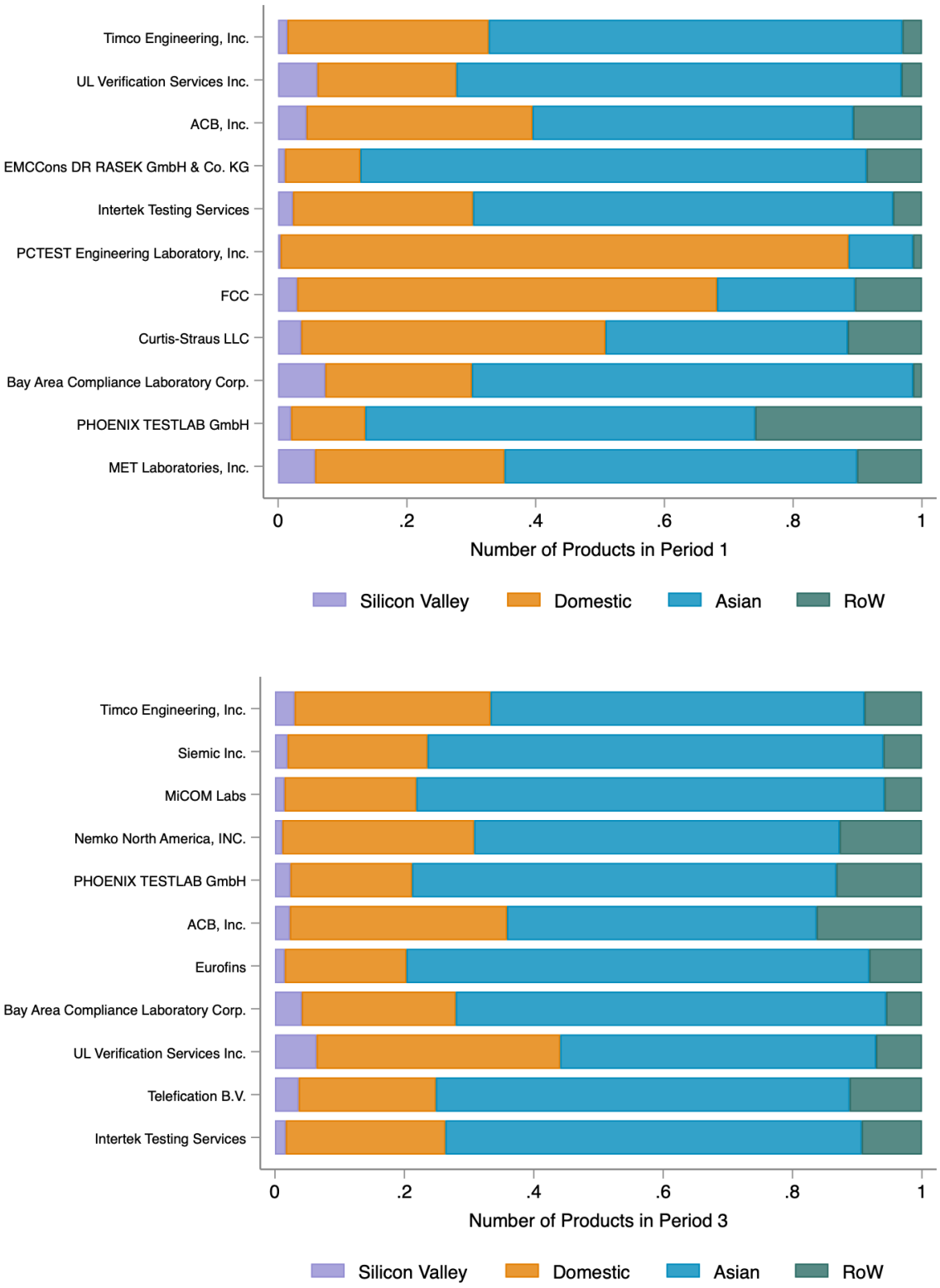


Figure A.3: Manufacturer Country Percent Breakdown for Top 10 TCBs by in 2001 to 2007 and 2011 to 2021. This figure plots the percent breakdown by geographic origin of certified products by the top ten Telecommunication Certification Bodies (TCBs). The top and bottom panels plot all products in 2008 to 2010 and 2011 to 2021, respectively. The purple bar represents Silicon Valley, the orange bar represents US/Canada (excluding Silicon Valley), the blue bar reppresents Asia, and the green bar represents Rest of World.